

NeuroChip Twin v2

A calibrated, data-driven digital twin of neural-network formation:
few-shot forecasting, hazard calls with fewer wells and next-dose planning,
validated on real microelectrode-array electrophysiology

Submission category: End-to-End System

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Built with two AI coding assistants used under the team lead's direction and declared as tools (Section 10).

Code and data bundle: github.com/Agnuxo1/neurochip-twin

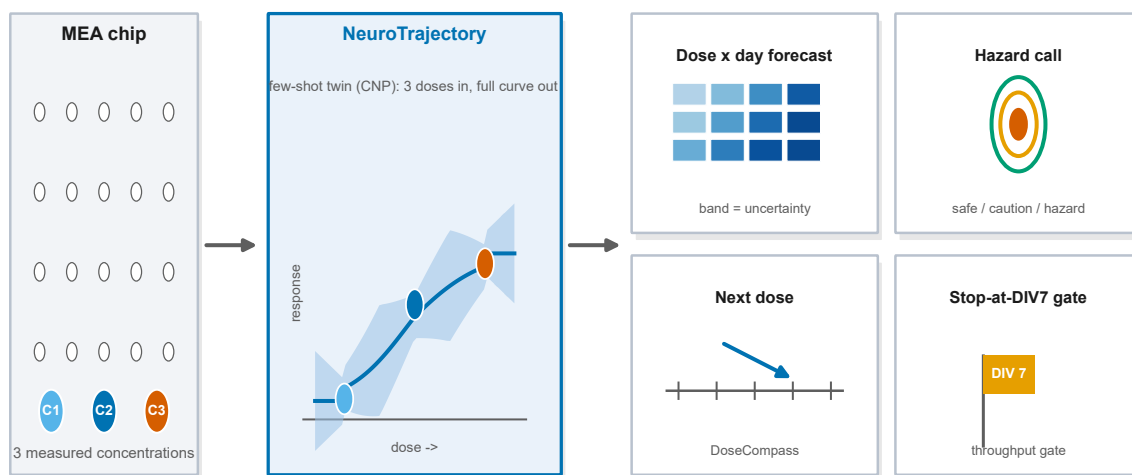
Kaggle notebook: kaggle.com/code/franciscoangulo

Interactive explorer: agnuxo1.github.io/neurochip-twin

Video (≤ 5 min): youtu.be/yLzpvGqP_2A

NeuroChip Twin v2

few-shot forecasting of neural network development on a chip



11%

89.9%

65 vs 55

26%

57%

lower forecast error at $k=3$

conformal coverage (target 90%)

neurotoxicants flagged, ours vs EPA rule

chemicals stop at DIV 7 (throughput gate)

median wells saved ($k=3$)

Summary. Every compound tested on a neural organ-on-a-chip or microelectrode-array (MEA) platform costs a full dose-response campaign: typically seven concentrations, replicate wells and twelve days of culture, analysed one endpoint at a time and forgotten by the next compound. NeuroChip Twin v2 learns from 243 chemicals of the US EPA network formation assay (rat cortical neurons, 17 network features at DIV 5, 7, 9 and 12). Given three measured concentrations of a new chemical, it forecasts the whole dose $\times$ day response with calibrated uncertainty, calls its developmental-neurotoxicity hazard, proposes the next concentration and flags cultures that can stop at DIV 7. Every result is out of fold, against executed baselines and an exact reproduction of EPA's hazard rules, and every number in this report is rendered from a versioned result file.

Table 1. Key numbers (unseen chemicals; 95% CIs in Table 4 and the Results).

Forecast from three doses	11.0% lower curve error than the best of five executed baselines (11.2% on four never-inspected folds)
Hazard call from three doses	65 known neurotoxicants detected against 52–56 for EPA’s rules on all concentrations (same 97 chemicals), with 57.1% fewer treated wells
Calibrated intervals	coverage 0.899 for nominal 90% intervals; abstained forecasts (10.7%, close to the rate fixed by the threshold) are the least reliable ones
Stop at DIV 7	26% of chemicals stop after two of four recordings (throughput gate); the twin’s safety gate lets 3 known neurotoxicants exit instead of 13
Next dose	measuring the top concentration first raises activity detection from 88.1% to 91.0%
Next assay to add	full-range viability would cut forecast error by 8.3% with one measured dose (scenario)

Innovation. (1) *A chemical is a few-shot task.* One model, meta-learned across 243 chemicals, forecasts the full dose $\times$ DIV development of all 17 network features for a chemical it has never seen (Section 5.1). (2) *An interpolation-informed gated decoder with dose-local attention,* which learns per feature and day how far to trust the measured neighbours; removing it costs 6.9% accuracy at $k = 3$ (Section 5.9). (3) *A toxicology-specific trust layer:* cross-conformal intervals stratified by cytotoxicity and number of measured doses, with epistemic abstention (Section 5.2). (4) *A closed decision loop for the lab:* hazard call, next concentration and a DIV 7 exit gate, each validated on real wells under preregistered protocols against exactly reproduced EPA rules (Sections 5.3, 5.5 and 5.6).

Reading guide. Table 15 maps every section to the Kaggle and Pazhou evaluation criteria. Evidence boundaries are stated in Section 6; tied and negative results, which shaped the design, are in Sections 5.4 to 5.6, 5.11 and 5.12 and Appendix A.

1 Problem and users

The bottleneck. Developmental neurotoxicity (DNT) testing is moving from animal studies to an in-vitro battery of new approach methodologies [1–3]. Its functional anchor is the network formation assay (NFA): rat primary cortical neurons grow on 48-well MEA plates while exposed to a chemical, and spontaneous activity is recorded as the network forms at days in vitro (DIV) 5, 7, 9 and 12 [4–6]. Every compound costs a full dose–response campaign: 212 of the 243 chemicals in the public NFA record were tested at seven concentrations, each in replicate wells followed to DIV 12. Compartmentalised neural chips, which impose directed axonal connectivity between sub-networks [7–9], raise the cost of each data point further. Analysis is still descriptive: one concentration–response curve per endpoint and day [10, 11], no calibrated uncertainty, and no use of the hundreds of compounds already measured. The NFA readout, spontaneous network activity on an MEA, is the same functional readout that compartmentalised neural chips use; the twin consumes any MEA recording mapped to its assay and chip descriptors, as shown on a four-compartment chip (Section 5.7) and in the sponsor-data path of Section 7.

Our question. *After measuring one to four concentrations of a new compound, can we forecast its full effect on neural-network development—every feature, every day, every concentration—with calibrated uncertainty, decide whether it is hazardous, and choose the next concentration to measure?* A model that answers this is a digital twin in the sense the challenge sponsor describes, “from experimental description toward predictive simulation”: a data-driven simulator of network formation that can be queried at conditions that were never run, and that is updated by every new experiment.

Users. (i) A neural organ-on-a-chip laboratory that runs compartmentalised chips instrumented for electrophysiology and wants to spend its wells and chips on informative conditions, and to turn its recordings into a reusable, predictive data asset. (ii) Regulatory and pharmaceutical toxicologists running the DNT in-vitro battery, who need hazard calls and potencies with honest uncertainty from fewer wells.

What the system delivers. (1) A provenance-tracked data asset (chemical $\times$ dose $\times$ DIV $\times$ well $\times$ feature) with cohort and splits frozen by hash (Section 2); (2) NeuroTrajectory, a few-shot forecast of the 4×17 developmental trajectory at any concentration for a chemical never seen in training (Sections 3 and 5.1); (3) calibrated intervals with visible abstention (Section 5.2), a hazard call from three concentrations (Section 5.3) and potency from the most reliable estimator we found (Section 5.4); (4) DoseCompass, the next concentration to measure, validated retrospectively on real wells (Section 5.5); and (5) a chip layer fitted on axon-resolved recordings of a four-compartment hippocampal chip (Section 5.7).

Table 2. Data sources. Counts are those of the frozen cohort after identity resolution. *Unpaired* means that no well, plate or culture is shared with the NFA data.

Source	Content used	Role	Licence / terms
EPA NFA [6, 11]	rat cortical networks, 48-well MEA, DIV 5/7/9/12, 17 features; 243 chemicals (267 sample IDs), 31,756 well $\times$ DIV records, 163 plates, 1,783 chemical $\times$ concentration levels	primary training and evaluation asset	US Government work; data only, analysis code not redistributed
EPA DNT reference labels [12, 13]	79 positive, 19 negative, 145 unlabelled chemicals in the cohort	hazard evaluation (R4, R5)	US Government work
EPA tcplfit2 NFA fits [10, 11]	published AUC and DIV12 endpoint fits for 241 chemicals	independent potency reference (R3); EPA-rule reproduction (R4)	US Government work
PubChem / invitrodb [14, 15]	AlamarBlue (metabolic viability) and LDH (membrane damage) calls of the NFA (AIDs 2284083, 2284068), chemical level	Mondrian strata (R7); R5	US Government work
NTP DNT-DIVER [16, 17]	well-level viability for 64 cohort chemicals	viability-join audit	US Government work
Brewer 4-compartment MEA [18–20]	EC $\rightarrow$ DG $\rightarrow$ CA3 $\rightarrow$ CA1 $\rightarrow$ EC sub-networks, 20 single-axon tunnels per array; 21 recordings, 440 sorted axons (202 feed-forward, 238 feedback)	chip layer (R8)	CC0 1.0
Harrill high-content imaging [21, 22]	74 chemicals, 21 neurite, synapse and viability endpoints, 192 plates	unpaired corroboration (X2)	US Government work
Kosnik acute MEA [23, 24]	384 samples on mature networks; literature truth set 41 neuroactive / 32 negative	unpaired contrast (X3)	US Government work

2 Data asset and compliance

Table 2 lists every data source. All are public and licensed for reuse; none contains personal or clinical data. Figure 1 summarises the primary asset.

Target representation. For feature f of a well recorded on plate p , date t and day d , the raw value x is variance-stabilised and expressed as an effect size in units of vehicle variability,

$$g_f(x) = \operatorname{asinh}(x/c_f), \quad y_{f,d} = \operatorname{clip}_{\pm 10} \left(\frac{g_f(x) - \operatorname{med}_{p,t,d}^{\text{veh}} g_f}{s_{f,d}} \right), \quad (1)$$

where c_f is a quarter of the pooled DIV 12 vehicle median, the median runs over the vehicle wells of the same plate, date and day, and $s_{f,d}$ is the pooled within-plate robust standard deviation of vehicle wells. All constants are fitted on vehicle wells only. Undefined values (for example burst duration in a well without bursts) are masked, never imputed. Wells with a failed quality flag are excluded.

Frozen cohort and splits. Before any model was fitted, the cohort, alias map, quality rules, normalisation and four split schemes were frozen and hashed (freeze G0, Appendix D). All sample IDs of a chemical, connected by DTXSID or normalised name, fall in the same fold of the primary five-fold chemical split, stratified by DNT label. Distinct substances are never merged (valproate and valproic acid, heptachlor and its epoxide are linked for leakage checks but kept apart). Secondary splits by plate, by held-out dose level and by chemical family are frozen as well; the results in this report use the primary chemical split only.

Compliance and licences. EPA data are used as data; the EPA analysis repositories carry no licence file, so their code is cited, not copied. Datasets with non-commercial terms were excluded (Appendix E). The R-data reader pyreadr (AGPL-3.0) was replaced by rdata (MIT) with byte-identical output, recorded as freeze amendment G0-A001. The code is released under the MIT licence.

What the feature layer is, and is not. We intended to validate a clean-room re-implementation of the 17 NFA features against EPA’s per-well values (R1). The public spike archive shares 0 plates with the NFA record

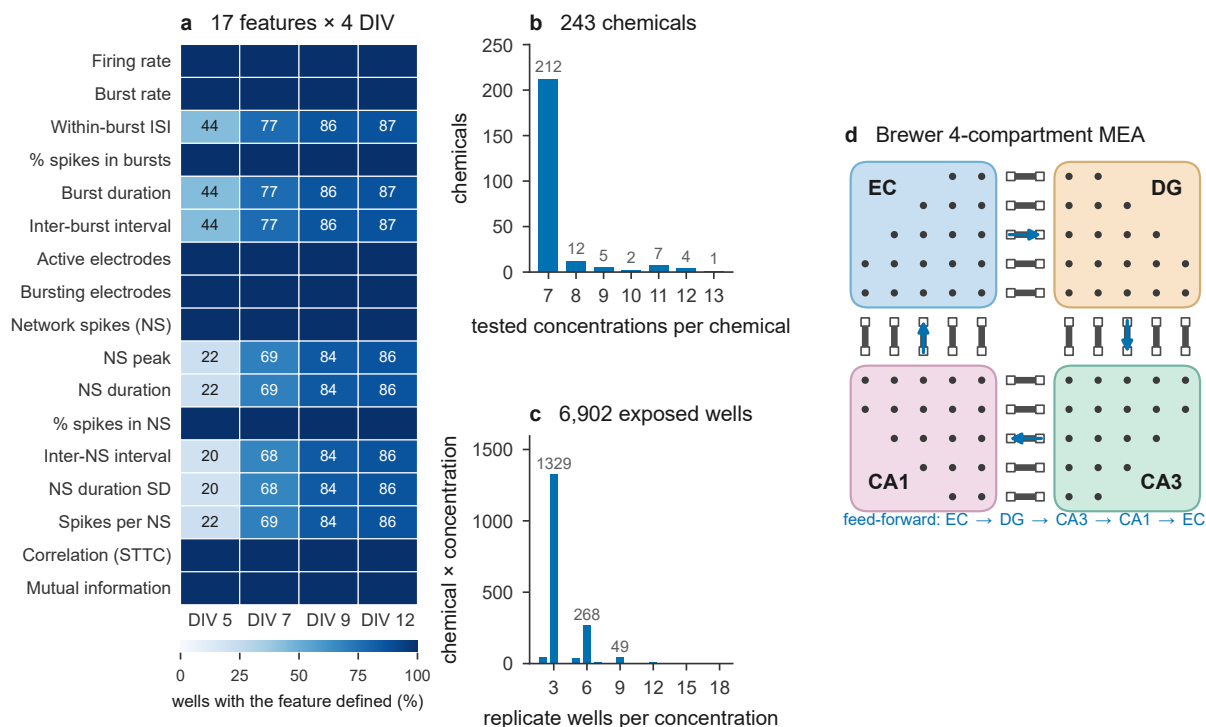


Figure 1. The data asset. (a) Tested concentrations per chemical and (b) replicate wells per chemical × concentration in the EPA NFA cohort. (c) Fraction of wells in which each feature is defined, by DIV; burst and network-spike features are undefined in young networks and are masked, not imputed. (d) Layout of the Brewer four-compartment MEA: four sub-networks joined by single-axon tunnels with an electrode at each end, which resolves the direction and conduction time of each axon.

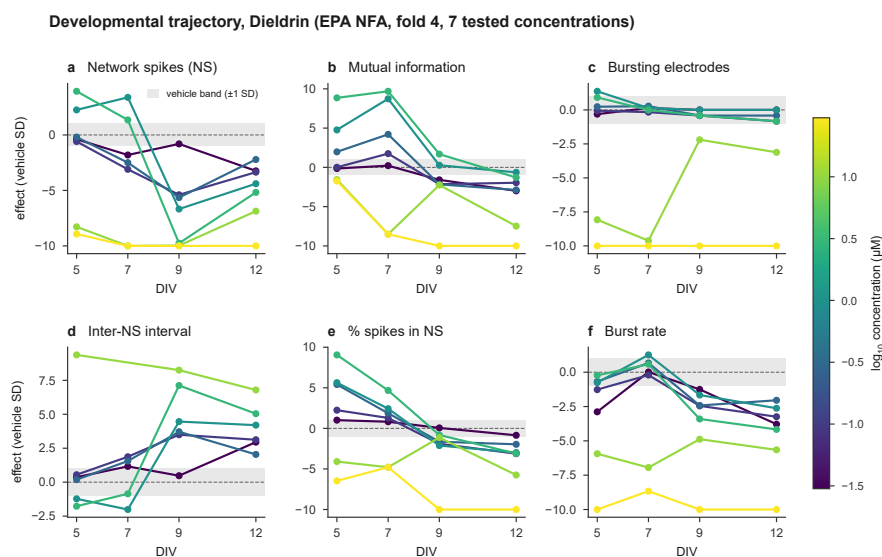


Figure 2. What the primary data look like: measured development (DIV 5–12) of six network features for dieltrin, the reference-positive chemical with the largest measured effect in folds 1–4 (selection rule in `make_visuals.py`); colour is concentration, grey is the vehicle band.

(9 archive plates versus 163), so fidelity is not estimable. The twin therefore models EPA’s published features, and our feature engine is applied to the Brewer chip only, as cross-platform analogues [25, 26].

3 Method

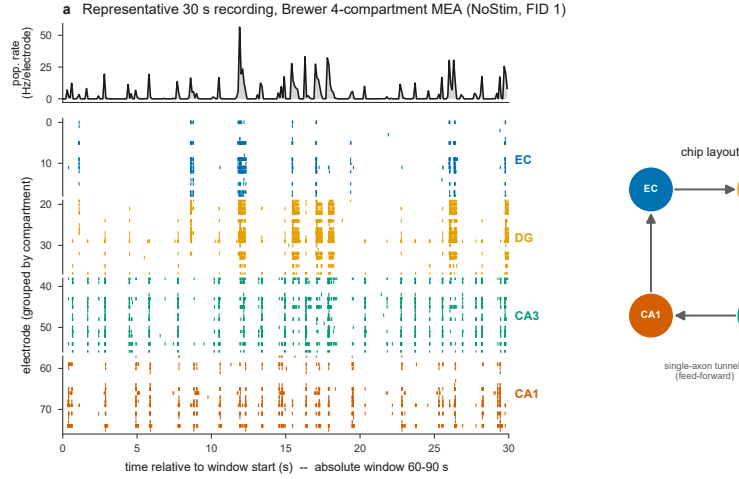


Figure 3. What the chip data look like: thirty seconds of spontaneous activity from an unstimulated four-compartment Brewer chip, electrodes grouped by sub-network, with the population rate above and the directed tunnel ring $EC \rightarrow DG \rightarrow CA3 \rightarrow CA1 \rightarrow EC$.

3.1 Formulation: a chemical is a few-shot task

For chemical c , the wells form a set $\mathcal{T}_c = \{(\ell_i, Y_i, M_i)\}_{i=1}^{n_c}$, where ℓ_i is the $\log_{10}$ concentration (μM), $Y_i \in \mathbb{R}^{4 \times 17}$ is the whole developmental trajectory of well i in the units of Equation (1), and M_i its observation mask. *Measuring k concentration levels reveals every replicate well at those levels*, the context C_k ; the task is to predict the wells at all other levels, and at any concentration in between, for a chemical never seen in training. $k = 0$ asks for the population prior.

3.2 NeuroTrajectory

NeuroTrajectory (Figure 4) is a conditional neural process [27, 28] with a dose-local attentive path [29, 30] and an interpolation-informed residual decoder. Concentrations enter through Fourier features [31], $e(\ell) = W_e[z, \sin(\omega z), \cos(\omega z)]$ with $z = (\ell + 0.5)/1.5$ and $\omega_j = 0.5 \cdot 2^j$, $j = 0, \dots, n_\omega - 1$; fewer frequencies make the response smoother in dose. For a query concentration q ,

$$h_i = \phi([e(\ell_i), Y_i \odot M_i, M_i]), \quad r_g = \frac{1}{|C_k|} \sum_{i \in C_k} h_i, \quad r_\ell(q) = \text{MHA}(W_q e(q), W_k e(\ell_i), h_i), \quad (2)$$

$$(\delta, s, \gamma) = \psi([r_g, r_\ell(q), e(q), W_I[I(q), \Delta(q)]]), \quad \mu(q) = \sigma(\gamma) \odot I(q) + \delta, \quad \sigma(q) = 0.02 + \text{softplus}(s), \quad (3)$$

where ϕ, ψ are multilayer perceptrons, MHA is multi-head attention whose keys are the measured doses, $I(q) \in \mathbb{R}^{4 \times 17}$ is the piecewise-linear interpolation in log dose of the context level means (flat outside the measured range), $\Delta(q)$ is the distance to the nearest measured level, and $\sigma(\cdot)$ is the logistic gate. The gate lets the network decide, per feature and day, how far to trust the measured neighbours and how much to correct them with what it has learned across chemicals. With an empty context, r_g and r_ℓ are learned vectors, $I \equiv 0$, and the model returns its prior.

Meta-training. Each episode samples a training chemical and a random number $k \leq 5$ of its levels as context, and scores the masked negative log-likelihood of all its wells under a Gaussian or a Laplace likelihood,

$$\mathcal{L} = -\frac{1}{\sum M} \sum_{i,d,f} M_{i,d,f} \log p(y_{i,d,f} | \mu_{d,f}(\ell_i), \sigma_{d,f}(\ell_i)), \quad (4)$$

with AdamW, a one-cycle schedule, 4000 steps and batches of 32 fixed a priori (PyTorch [32]).

Nested selection and ensemble. Inside each outer training set, one inner fold chooses the likelihood (Gaussian or Laplace) and the decoder (plain or interpolation-informed); outer test chemicals are never used for any choice. The interpolation-informed decoder was selected in every outer fold. The chosen configuration is re-trained with three seeds, and the ensemble predicts $\bar{\mu} = \frac{1}{3} \sum_m \mu_m$ with scale $\bar{\sigma}^2 = \frac{1}{3} \sum_m \sigma_m^2 + \text{Var}_m(\mu_m)$. A second grid (v2) adds the number of dose frequencies $n_\omega \in \{3, 8\}$ as a smoothness hyperparameter; we report it as a robustness check.

3.3 Calibration and abstention

Every prediction used for calibration is out of fold. For a held-out well entry the nonconformity score is $s = |y - \bar{\mu}|/\bar{\sigma}$. In cross-conformal fashion [33–36], the interval for outer fold g and Mondrian stratum π (cytotoxicity regime $\times$ number of measured levels k) is

$$\bar{\mu} \pm \hat{q}_{g,\pi} \bar{\sigma}, \quad \hat{q}_{g,\pi} = \text{the } [(n+1)(1-\alpha)]/n \text{ empirical quantile of } \{s : \text{fold} \neq g, \text{stratum} = \pi\}. \quad (5)$$

The cytotoxicity regime is set at chemical level from the AlamarBlue and LDH calls of the NFA assay (*cytotoxic* if either is active, *non-cytotoxic* if both are inactive, otherwise *indeterminate*); it is never a same-well viability readout. The twin abstains when the epistemic ratio $e = \overline{\text{sd}_m(\mu_m)}/\bar{\sigma}$ exceeds the 90th percentile of its values in the calibration folds.

3.4 Derived potency and hazard

The developmental summary of feature f is $A_f(\ell) = \frac{1}{4} \sum_d y_{f,d}(\ell)$, in vehicle standard deviations. A chemical is active on f if $|A_f|$ reaches the benchmark response $\text{BMR} = 3$ within the tested range; $\text{BMC}_f = \min\{\ell : |A_f(\ell)| \geq \text{BMR}\}$ on a 120-point log grid, and the chemical’s potency is its most sensitive feature, $\text{BMC} = \min_f \text{BMC}_f$. This BMC is a threshold-crossing concentration on a grid, not a model-fitted benchmark dose (BMD/BMDL) such as those of EPA’s tcplfit2; comparisons with tcplfit2 potencies (R3) are rank and error comparisons, not equivalence. The hazard call is an L2 logistic regression (balanced class weights, $C = 1$ fixed a priori) on 51 trajectory summaries—for each feature, $\max |A_f|$, A_f at the top concentration and BMC_f —with out-of-fold probabilities from the frozen chemical folds.

3.5 DoseCompass: the next concentration by expected information

Given the measured levels, DoseCompass draws $S = 2000$ plausible full-information data sets: measured levels keep their observed summaries, and every unmeasured level ℓ receives $A_s(\ell) = \mu_A^{(m)}(\ell) + \tau \text{sd}_A^{(m)}(\ell) \varepsilon_s(\ell)$, where m cycles over ensemble members and ε_s is a Gaussian process in log dose with length scale ℓ_{GP} . Each sample yields a potency P_s (inactive, or the BMC in 0.25-log bins) and, for each candidate level u , an outcome $O_s(u)$ ($\max_f |A_s(u, f)|$ binned at 0.5, 1 and $2 \times \text{BMR}$). The next level maximises the expected information gain [37, 38]

$$u^* = \arg \max_u \text{EIG}(u), \quad \text{EIG}(u) = I(P; O(u)) = \sum_{P,O} \hat{p}(P, O) \log_2 \frac{\hat{p}(P, O)}{\hat{p}(P) \hat{p}(O)}. \quad (6)$$

(τ, ℓ_{GP}) are chosen per outer fold by cross-fitting the log score of the reference potency bin on the other four folds only; the choice was $(\tau, \ell_{\text{GP}}) = (0.5, 1)$ in every fold, inside the grid. The planner reads only wells at measured levels.

3.6 ChipLayer: directed propagation on a four-compartment chip

For each recording, the four sub-networks form a directed ring ($\text{EC} \rightarrow \text{DG} \rightarrow \text{CA3} \rightarrow \text{CA1} \rightarrow \text{EC}$ is feed-forward); each tunnel carries sorted axons with direction and conduction time [8, 19]. An empirical-Bayes model shrinks the feed-forward fraction of a tunnel towards its anatomical pair and the pair towards the global rate,

$$\hat{p}_{\text{pair}} = \frac{n_{\text{pair}}^{\text{ff}} + \kappa \hat{p}_0}{n_{\text{pair}} + \kappa}, \quad \hat{p}_t = \frac{n_t^{\text{ff}} + \kappa \hat{p}_{\text{pair}(t)}}{n_t + \kappa}, \quad \kappa = 8 \text{ axons}, \quad (7)$$

with the same shrinkage for the discrete distribution of conduction times. Because the orientation of an array changes which anatomical pair a physical tunnel joins, pairs are resolved per recording.

3.7 Baselines and statistics

Every method receives exactly the same measured wells: zero effect; the mean of the measured levels; piecewise-linear interpolation in log dose; a Hill curve per endpoint (exhaustive search over AC50 and slope, closed-form efficacy); and analog nearest-neighbour transfer from the training chemicals whose responses at the measured concentrations are closest. The primary forecasting metric is the *curve MAE*: the absolute error against the mean of the replicate wells at each held-out level, averaged per chemical and then over chemicals, over five seeded designs per k . Differences are paired by chemical and bootstrapped over chemicals (4000 resamples) [39]; proportions carry Clopper–Pearson intervals [40]; paired classifiers are compared with the exact McNemar test [41].

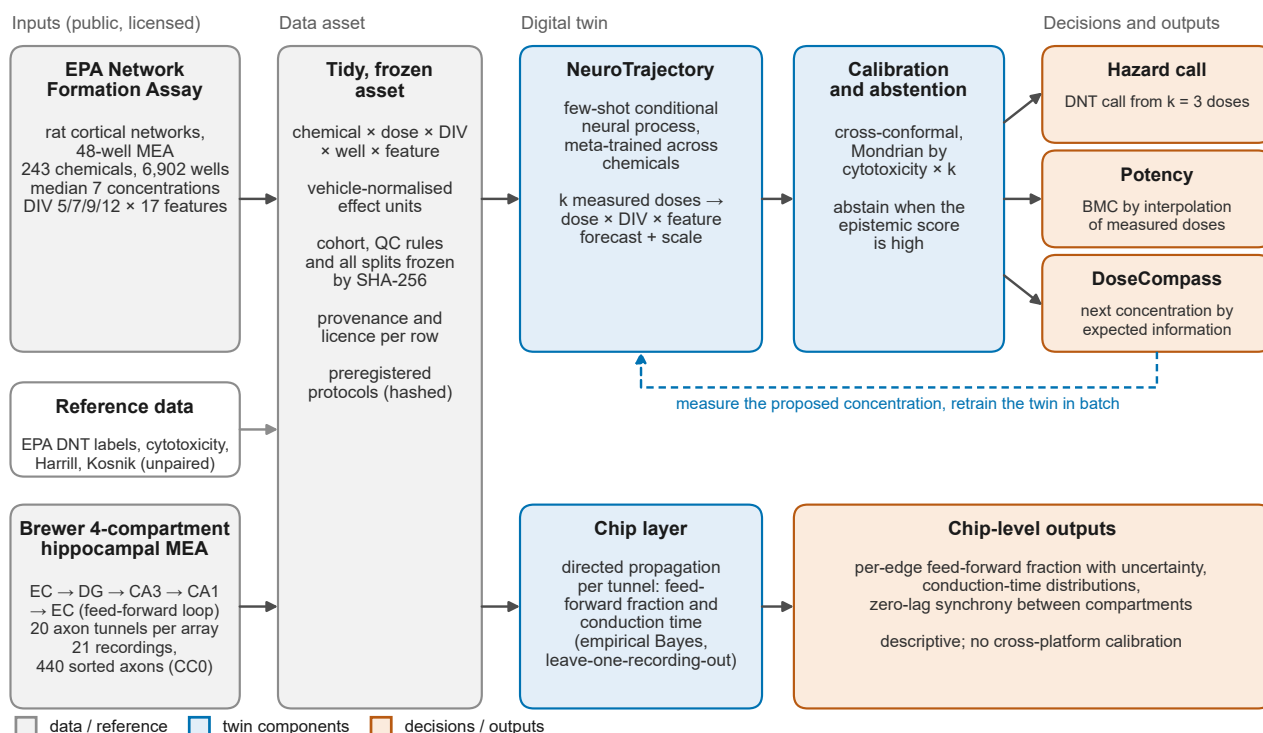


Figure 4. System overview. Public, licensed inputs are unified into a frozen, provenance-tracked asset. NeuroTrajectory forecasts the dose × DIV × feature response of an unseen chemical from k measured concentrations; calibration and abstention wrap every forecast; the decision layer returns a hazard call, a potency and the next concentration to measure, closing the laboratory loop. The chip layer is fitted on axon-resolved recordings of a four-compartment hippocampal chip and is descriptive (no cross-platform calibration).

Table 3. Modules, entry points and the result files they write.

Layer	Code (src/neurotwin/, scripts/)	Output (results/)
Ingestion and freeze	data/epa_nfa.py, data/cohort.py, data/brewer.py, data/harrill.py, data/kosnik.py	freeze_g0.json, cohort_freeze.json, splits_*.json, data_audit_brewer.json
EPA-rule reproduction	data/epa_nfa.py (hit-count rules on EPA's published fits)	epa_baseline_repro.json
Forecasting (NeuroTrajectory)	models/cnp.py, models/trajectory.py, scripts/run_trajectory_cv.py	trajectory_cv.json, trajectory_cv_v2.json
Calibration, abstention	scripts/run_conformal.py	conformal_r7.json
Potency and hazard	models/potency.py, scripts/run_potency.py, scripts/run_dnt_r4.py	potency_r3.json, potency_r3b.json, dnt_r4.json
Planning (DoseCompass)	planner/eig.py, scripts/run_dosecompass*.py	r6_dosecompass.json, r6b_dosecompass_anchored.json
Chip layer	chip/propagation.py, features/mea.py, scripts/run_chiplayer.py	r8_chiplayer.json, chiplayer_params.json, brewer_features.json
Cross-assay, diagnostics	scripts/run_crossassay.py, run_ablations.py, run_selectivity_failures.py	x2_harrill.json, x3_kosnik.json, r9_ablations.json, r10_failures.json, r5_selectivity.json

4 System and implementation

The system is a Python package (src/neurotwin) with one module per layer and a script per result; each script writes one JSON file in results/, and nothing else is a source of numbers (Table 3).

Delivery. Four entry points share one compute core and the cross-validated weights shipped in the data bundle. The command-line demo runs on a laptop CPU without network access, loads the models of the fold that never saw the chosen chemical, forecasts its trajectory from k measured concentrations and then reveals the held-out wells and scores the forecast against the same baselines; a Gradio app exposes the same core

Table 4. Results at a glance (primary analyses; 95% confidence intervals in brackets).

ID	Question	Result	Verdict
R2	Forecast unseen chemicals from $k = 3$ doses?	curve MAE 1.180 vs 1.325 (best baseline); -0.145 $[-0.188, -0.099]$; folds 1–4: 11.2% lower	positive
R7	Are 90% intervals calibrated?	coverage 0.899 $[0.891, 0.906]$ after conformal (0.882 before); abstention 10.7%	positive
R4	Hazard call from three doses?	BA 86.4% vs 82.6–85.9% (EPA rules, same 97 chemicals); McNemar significant in one of three	modest
R3	Potency from the model?	model-derived BMC worse than interpolation of measured doses at every k	negative
R6	Choose the next dose?	ties log-spaced $(-0.006$ $[-0.060, 0.053])$; 9.1% better than random	tie
R8	Directed chip connectivity?	feed-forward fraction MAE -0.036 vs tunnel means; conduction: no gain	modest
X2	Network vs morphology?	NFA active in 96% of morphology-active chemicals; network BMC lower by $0.56 \log_{10}$	positive
X3	Developmental vs acute?	activity BA 65.4%; potency ranks do not transfer	modest
R9	Do the components matter?	removing the interpolation-informed decoder: +6.9% error at $k = 3$	positive
R5	Specific vs cytotoxic disruption?	selectivity index AUROC 0.51, no better than chance	negative

Table 5. R2: curve MAE (vehicle SD units; lower is better) for held-out chemicals, all five folds. Δ is NeuroTrajectory minus the best baseline at that k (marked †), paired over chemicals.

k	NeuroTrajectory	Interp.	Hill	Analog kNN	Ctx. mean	Zero	Δ vs best [95% CI]	$\Delta\%$ all	$\Delta\%$ folds 1–4
0	1.689	1.769	1.769	1.702 †	1.769	1.769	-0.013 $[-0.027, 0.000]$	−0.8	−0.4
1	1.415	1.875	1.875	1.494 †	1.875	1.770	-0.080 $[-0.115, -0.040]$	−5.3	−5.8
2	1.279	1.533	1.533	1.393 †	1.697	1.773	-0.114 $[-0.147, -0.078]$	−8.2	−8.8
3	1.180	1.325 †	1.435	1.327	1.607	1.783	-0.145 $[-0.188, -0.099]$	−11.0	−11.2
4	1.138	1.239 †	1.308	1.311	1.590	1.810	-0.100 $[-0.136, -0.061]$	−8.1	−8.8

interactively; a static explorer (GitHub Pages) serves precomputed held-out forecasts for every chemical; and a Kaggle notebook reproduces the headline tables. Training ran on a single consumer GPU; inference, the demo and the tests run on CPU.

Credibility layer. `scripts/render_docs.py` fills every number of this report, the README and the Writeup from `results/*.json`, stops when a token does not resolve, and writes a ledger of each value with its source; `scripts/check_numbers.py` fails continuous integration when a rendered document (this report, its figure captions, the README) contains an untraced number, or when a template contains a typed number that is not a structural constant justified in `docs/number_allowlist.txt`; `scripts/check_freeze_g0.py` verifies every frozen file against its hash.

5 Results

All results are out of fold under the frozen chemical split: a chemical is only ever scored by models that never saw it. Table 4 summarises them; each subsection states the protocol and our reading of the evidence, negative results included.

5.1 R2 — Forecasting network formation in unseen chemicals

Protocol. Five outer folds of 243 chemicals; for each held-out chemical and $k \in \{0, \dots, 4\}$, five seeded designs choose the measured levels; all other levels are targets. One short development smoke run (one seed) was evaluated on outer fold 0 and motivated adding the interpolation-informed decoder as a candidate; the final choice is nested, and we report folds 1–4 (194 chemicals, never inspected) separately.

Result. NeuroTrajectory has the lowest error at every k (Table 5 and Figure 5). With three measured concentrations its curve MAE is 1.180 against 1.325 for log-linear interpolation, the best baseline: a paired difference

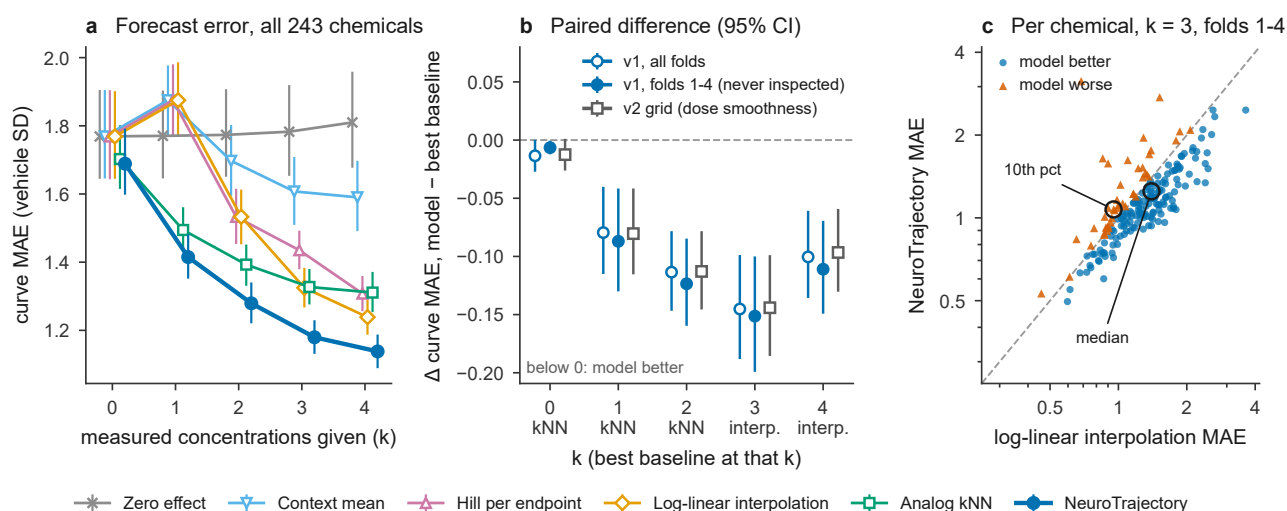


Figure 5. R2 forecasting. (a) Curve MAE of held-out chemicals against the number of measured concentrations k , with chemical-bootstrap 95% intervals. (b) Paired difference to the best baseline at each k for all folds, for folds 1–4 (never inspected during development) and for the v2 grid with tuned dose smoothness. (c) Per-chemical error at $k = 3$ on folds 1–4; points below the diagonal are chemicals where NeuroTrajectory is better.

Table 6. R7: observed coverage of nominal 90% intervals on held-out wells, before (parametric ensemble) and after cross-conformal calibration, with mean interval width (vehicle SD units) and coverage per Mondrian stratum.

k	Coverage (nominal 0.90)		Mean width		Conformal coverage by regime		
	parametric	conformal [95% CI]	param.	conf.	non-cytotoxic	cytotoxic	indeterminate
1	0.869	0.898 [0.890, 0.906]	7.26	8.32	0.899	0.896	0.903
2	0.875	0.899 [0.891, 0.907]	6.95	7.83	0.897	0.900	0.907
3	0.882	0.899 [0.891, 0.906]	6.77	7.41	0.897	0.900	0.904

of -0.145 [-0.188 , -0.099], 11.0% lower error, and a gain for 77% of chemicals. On the four never-inspected folds the gain is 11.2% (paired difference [-0.199 , -0.100]), for 77.3% of chemicals. The advantage grows from one to three measured concentrations and is still clear at four; with no measurement ($k = 0$) the learned prior only ties the analog baseline. Against a Hill curve per endpoint, the usual practice, the error falls by 17.8% at $k = 3$. The same ranking holds for the noisier well-level error (-0.121 [-0.148 , -0.092] against analog kNN at $k = 3$). The v2 grid, which also tunes dose smoothness, gives 1.181 at $k = 3$ (10.9% below the best baseline; 11.2% on folds 1–4): the result does not hinge on that hyperparameter.

Reading. Meta-learning across chemicals transfers: a new compound’s trajectory is predicted better by a model that has seen how 243 others perturb network formation than by any per-chemical fit of the same wells. The gain is largest where interpolation is weakest—between and beyond the measured doses, and in features whose response is not monotone in dose (Figure 6). Section 5.10 shows where the model loses.

5.2 R7 — Calibrated uncertainty and abstention

Protocol. Cross-conformal Mondrian calibration (Equation (5)) over the 243 out-of-fold chemicals (117 non-cytotoxic, 107 cytotoxic, 19 indeterminate), for $k = 1, 2, 3$ and nominal 80, 90 and 95% intervals. Coverage is the mean over held-out chemicals, with a chemical-bootstrap interval.

Result. The ensemble’s own intervals are slightly too narrow (0.882 coverage at $k = 3$ for a nominal 0.90). After calibration, coverage is 0.899 [0.891, 0.906] and is on target at every k , in every stratum and at every level: at $k = 3$, 0.800 for nominal 0.80 and 0.949 for nominal 0.95 (Table 6 and Figure 7). Calibration has a price that is itself informative: intervals are wider for cytotoxic than for non-cytotoxic chemicals (8.05 vs 6.66 vehicle SD at $k = 3$; indeterminate chemicals are as wide), because network responses there collapse abruptly, and all intervals narrow as doses are added. The abstention rule flags 10.7% of forecasts; their curve MAE is 1.666, against 1.250 for retained forecasts, and the abstention rate falls from 14.4% at $k = 1$ to 7.8% at $k = 3$.

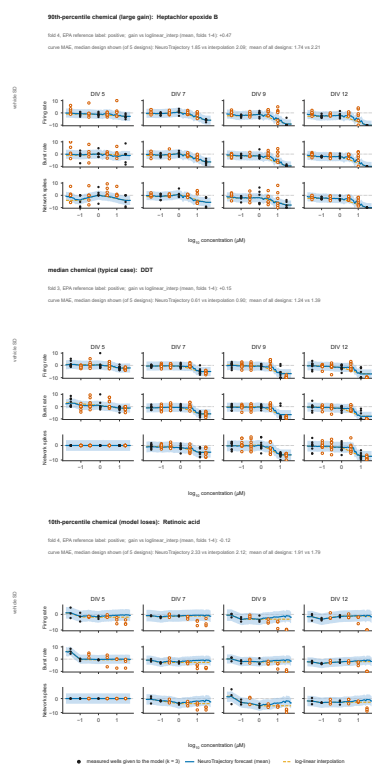


Figure 6. Held-out forecasts at $k = 3$ for three chemicals chosen by a fixed rule: the 90th, 50th and 10th percentiles of the per-chemical gain over interpolation (Heptachlor epoxide B, DDT, Retinoic acid; all reference positives), each with its median design, never its worst; rows are three fixed features. Black: wells shown to the model; open: held-out wells, revealed after forecasting. Blue: NeuroTrajectory mean with its pooled conformal 90% band; dashed: log-linear interpolation of the measured levels.

Table 7. R4: hazard calls on the same 97 reference chemicals (Clopper–Pearson 95% intervals). McNemar columns compare the pre-specified few-shot variant with each EPA rule: chemicals only we call correctly, only the rule calls correctly, and the exact two-sided p .

Method	Sensitivity (%)	Specificity (%)	BA (%)	ours only	rule only	p
NeuroTrajectory, $k = 3$ (pre-specified)	83.3 [73.2, 90.8]	89.5 [66.9, 98.7]	86.4			
NeuroTrajectory, all doses	80.8 [70.3, 88.8]	68.4 [43.5, 87.4]	74.6			
EPA rule DIV12	70.5 [59.1, 80.3]	94.7 [74.0, 99.9]	82.6	14	5	0.064
EPA rule AUC 3-hit	66.7 [55.1, 76.9]	100.0 [82.3, 100.0]	83.3	14	3	0.013
EPA rule Top-2	71.8 [60.5, 81.4]	100.0 [82.3, 100.0]	85.9	11	4	0.118

Reading. Coverage is a measured property inside the calibration domain (rat cortical NFA, chemical-level exchangeability), not a guarantee that transfers to other species, cell sources or chips. Within that domain, a user can read a twin interval at face value and can see when the twin declines to answer.

5.3 R4 — A hazard call from three concentrations

Protocol. Reference chemicals: the intersection of the EPA reference table with the cohort (79 positive, 19 negative). Two variants were declared before evaluation: *few-shot* ($k = 3$), whose 51 summaries come from the NeuroTrajectory forecast given three spread concentrations, and *all doses*, from the measured levels. Comparators are EPA’s published hit-count rules (DIV12, AUC 3-hit, Top-2), which we reproduce exactly on EPA’s own cohort: 86 positive and 19 negative chemicals, a difference of 0.0 percentage points on every selected published metric, and zero mismatches in the per-chemical hit sums. The comparison is paired on the 97 chemicals with a per-chemical EPA call. Two caveats are ours to state: EPA’s endpoint fits are taken as published, not refitted from wells, and its Top- k features were ranked on the full reference set (in sample), so its published balanced accuracy is not an out-of-sample ceiling.

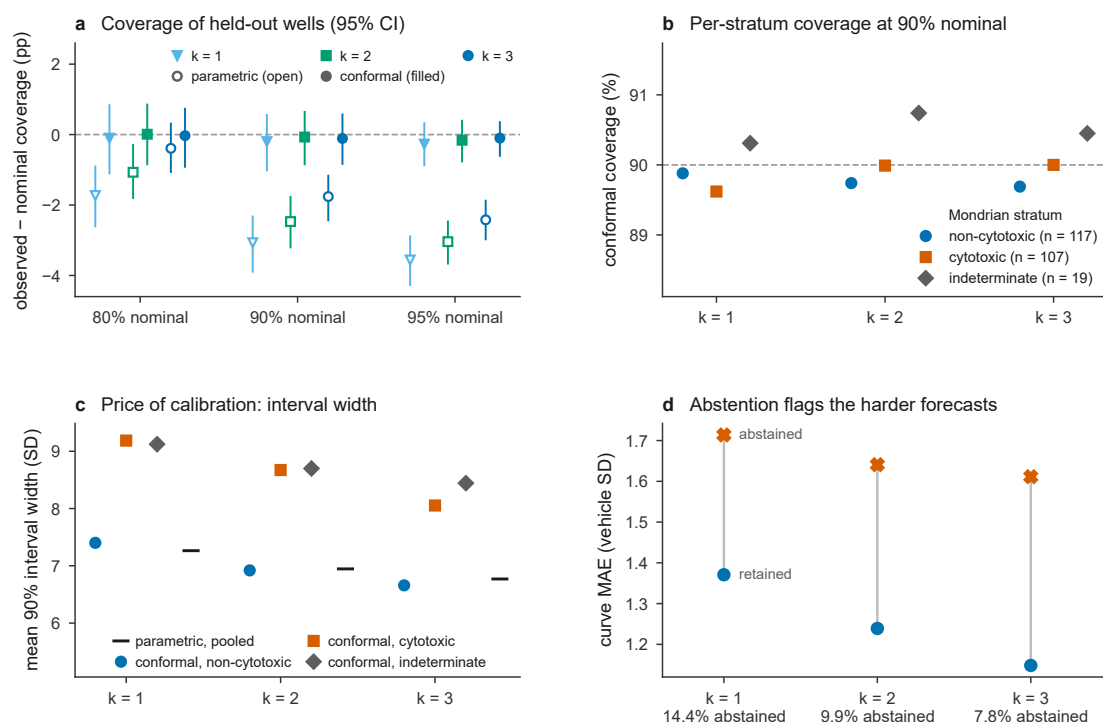


Figure 7. R7 calibration. (a) Observed minus nominal coverage of held-out wells, before (open) and after (filled) cross-conformal calibration, for $k = 1, 2, 3$. (b) Conformal coverage per Mondrian stratum at 90% nominal. (c) Mean width of 90% intervals: the price of calibration is paid in cytotoxic and indeterminate chemicals. (d) Curve MAE of retained and abstained forecasts.

Result. On the 97 chemicals with a per-chemical EPA call, three concentrations completed by the twin detect 65 known neurotoxicants, against 52–56 for EPA’s rules on the full series, at the cost of 2 false alarms among the negatives (EPA: 0–1); balanced accuracy is 86.4% (Table 7 and Figure 8). On all 79 + 19 reference chemicals the AUROC is 0.915 [0.853, 0.965]. Against AUC 3-hit, it calls 14 chemicals correctly that the rule misses and misses 3 that the rule gets right ($p = 0.013$); against DIV12 and Top-2 the differences are not significant. The all-doses variant is weaker (balanced accuracy 74.6%, McNemar $p \geq 0.36$ against every rule).

Reading. For a screening tier, the higher sensitivity is the more useful operating point: three concentrations, completed by the twin, carry at least as much hazard information as EPA’s rules applied to the full series. Cytotoxicity does not explain the result: most reference positives are cytotoxic and almost no negatives are (Section 5.11), but among the 53 non-cytotoxic chemicals (a secondary, post-hoc stratification) the twin’s balanced accuracy is 78.7%, against 66.2–72.1% for EPA’s rules, with lower sensitivity for every method. We do not claim superiority: with 19 negatives a specificity difference of one or two chemicals moves balanced accuracy by several points, only one of three pre-declared comparisons reaches $p < 0.05$, and EPA’s Top- k rules had an in-sample advantage. That the few-shot variant beats the all-doses variant is consistent with the forecast acting as a learned denoiser of replicate noise; we report it and do not build on it.

5.4 R3 and R3b — Potency: a negative result that set a design decision

Protocol. Potency (Section 3.4) recovered from $k = 2, 3, 4$ measured concentrations of the half of each chemical’s wells shown to the methods, scored against two independent references: EPA’s published tcplfit2 fits (activity agreement, and Spearman correlation of potencies for chemicals active in both), and a replicate-split reference computed from the other half of the wells at all levels, so that no noise is shared. The per-chemical composite error is $|\Delta\text{BMC}|$ ($\log_{10}$) if active in both, 1 on activity disagreement and 0 if both are inactive. R3b, registered in the decision log before it was run, completes the measured series with the forecast only at the tested but unmeasured levels.

Result. Potency read from the model’s dense-grid forecast is worse than interpolation of the measured doses at every k : by 8.7, 12.3 and 8.6% in composite error for $k = 2, 3, 4$, with intervals that exclude zero (Table 8, Figure 8c). The registered R3b estimator improves on the dense grid at $k = 3$ (-0.0365 [$-0.0744, -0.0002$]) and

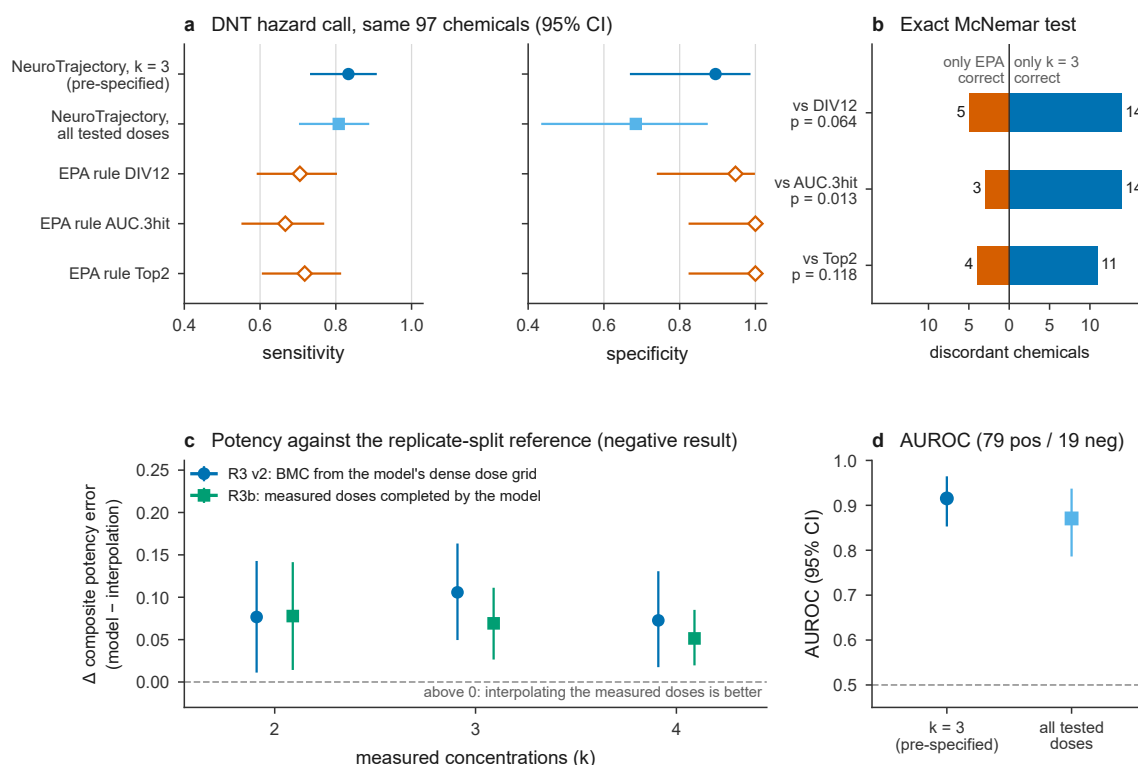


Figure 8. R4 hazard call and R3 potency. (a) Sensitivity and specificity with Clopper–Pearson 95% intervals on the same 97 reference chemicals, for NeuroTrajectory ($k = 3$, pre-specified; and all doses) and the reproduced EPA rules. (b) Discordant chemicals between the $k = 3$ call and each EPA rule, with the exact McNemar p . (c) R3/R3b potency: paired difference in composite error to interpolation of the measured doses (above zero, interpolation is better; this is why BMC is reported by interpolation).

Table 8. R3/R3b: potency error against the replicate-split reference (composite error, lower is better) and rank agreement with EPA tcplfit2 potencies at $k = 3$. The last column is the paired difference to log-linear interpolation at $k = 3$ (positive = worse than interpolation).

Estimator	Composite error, replicate split			Spearman vs EPA $k = 3$	Δ vs interp., $k = 3$ [95% CI]
	$k = 2$	$k = 3$	$k = 4$		
NeuroTrajectory, dense grid (R3)	0.960	0.967	0.920	0.65	0.106 [0.050, 0.163]
NeuroTrajectory, completed series (R3b)	0.961	0.930	0.898	0.66	0.069 [0.026, 0.111]
Log-linear interpolation	0.884	0.861	0.847	0.71	—
Hill per endpoint	0.884	0.898	0.857	0.78	
Analog kNN	1.039	0.968	0.946	0.74	

has the best activity agreement with the replicate reference (Cohen’s κ of 0.27 versus 0.21), but it still loses to interpolation in composite error. Against EPA’s fits, a Hill curve per endpoint ranks potencies best at $k = 3$. For scale, our potency rule applied to all tested levels agrees with EPA’s activity calls with $\kappa = 0.67$ and ranks potencies with Spearman 0.75.

Reading and decision. A benchmark crossing is a threshold event; small ripples of a smooth forecast between tested concentrations create spurious early crossings, while interpolation cannot cross between two sub-threshold measurements. We therefore *report BMC by interpolation of measured doses* and use the twin for what it does well—the trajectory, its uncertainty and the hazard call. A first version of this protocol (kept as `potency_r3_v1_superseded_biased_reference.json`) used as reference the interpolation of all level means, which coincides with the interpolation estimator at the measured levels and therefore favoured it (under that reference the model appeared 26.2% worse than interpolation at $k = 3$). It was replaced by the two independent references above; the gap shrank, but the conclusion did not change.

Table 9. R6/R6b at $B = 3$ (NeuroTrajectory estimator). Composite error (lower is better), activity accuracy and median $|\Delta\text{BMC}|$ among chemicals active in both; Δ is DoseCompass minus the comparator, paired over the 243 chemicals.

Protocol	Strategy	Composite	Activity acc.	Med. $ \Delta\text{BMC} $	Δ [95% CI]
R6 (median first)	DoseCompass	0.445	0.840	0.221	
	fixed log-spaced	0.451	0.881	0.287	−0.006 [−0.060, 0.053]
	random (mean of 200)	0.489	0.840	0.260	−0.045 [−0.088, −0.002]
	DoseCompass, DIV 5–9 only	0.444	0.852	0.244	
	oracle (upper bound)	0.147	0.947	0.037	
R6b (top first)	DoseCompass, anchored	0.432	0.909	0.323	
	fixed log-spaced	0.451	0.881	0.287	−0.019 [−0.073, 0.033]
	random, anchored	0.456	0.900	0.297	−0.024 [−0.065, 0.013]

5.5 R6 and R6b – Choosing the next concentration

Protocol. Retrospective and preregistered (protocol SHA-256 in Appendix D): with a budget of $B = 3$ levels, every strategy starts from the same first level—the median tested concentration in R6 and the highest in R6b—and adds $B - 1$ levels. DoseCompass is compared with fixed log-spaced designs (primary), with 200 seeded random orders and with an oracle subset that uses the hidden wells (an upper bound, not a strategy). Potency is then estimated from the chosen levels by NeuroTrajectory; the metric is the composite error of R3 against the full-information reference. R6b, the second protocol, was motivated by R6 (omitting the top concentration cost activity detection); both are reported, without multiplicity correction.

Result. On the primary endpoint DoseCompass ties fixed log-spaced designs in both protocols (Table 9 and Figure 9). It beats random selection by 9.1% in R6, with an interval that just excludes zero, and not significantly in R6b. Anchoring on the top concentration raises activity detection: in R6b, DoseCompass calls activity correctly for 91.0% of chemicals ($\kappa = 0.76$) against 88.1% ($\kappa = 0.68$) for log-spaced designs, and it cuts the composite error on reference-inactive chemicals from 0.250 to 0.183. The planner’s belief is informative: after three levels its probability of activity has a Brier score of 0.104, against 0.186 for the base rate. Deciding with DIV 5–9 only, before the last recording, loses nothing (−0.001 [−0.044, 0.045]).

Reading. On EPA’s grid of pre-chosen concentrations (seven for most chemicals), a good fixed design is hard to beat, and the large gap to the oracle is mostly irreducible without seeing the wells. The operational lessons are concrete: measure the top concentration first, then let the planner place the remaining budget, and decide before DIV 12 if time matters. A prospective trial with a free choice of concentrations is where the planner can show more (Section 7).

5.6 R11 – Early exit: stop the culture at DIV 7

Protocol. A neural culture takes twelve days and four recordings. R11 asks whether the first two recordings (DIV 5 and 7) are enough to decide which chemicals the complete assay would call inactive, so that their culture and recordings can stop early while (almost) every active chemical continues. The idea comes from the team lead’s early-rejection filter for hardware pipelines, which discards work items from an early partial signal before full processing [42]. Here it is tested on biology with a preregistered protocol (Appendix D). The cohort is the 194 unseen chemicals of folds 1–4. The reference is the fixed activity rule of Section 3.4 applied to all DIVs and concentrations, which gives 146 active and 48 inactive chemicals. Four scores see DIV 5/7 only: NeuroTrajectory with DIV 9/12 masked, the same forecast with its upper 90% band, the rule applied to the early data, and persistence (DIV 7 carried forward). For each fold, the exit threshold keeps 95% of the active chemicals of the other three folds. The primary endpoint is the fraction of inactive chemicals that exit.

Result. Early exit works, and it has two operating points (Table 10). Against the assay’s own rule, persistence is the most efficient gate: it stops 92% of the inactive chemicals at DIV 7 while 95% of the active ones continue. NeuroTrajectory, trained only with full-trajectory contexts, stops 46% (difference −0.46 [−0.62, −0.30]). By the preregistered rule, persistence is the recommended throughput gate. An independent verifier recomputed every score. The reference shares the DIV 5/7 wells with every score, so R11b also evaluates every method against a reference built from DIV 9/12 only. The external EPA DNT labels reverse the ranking where it matters most for safety: persistence would stop 13 of 63 known neurotoxicants at DIV 7, the twin’s upper 90%

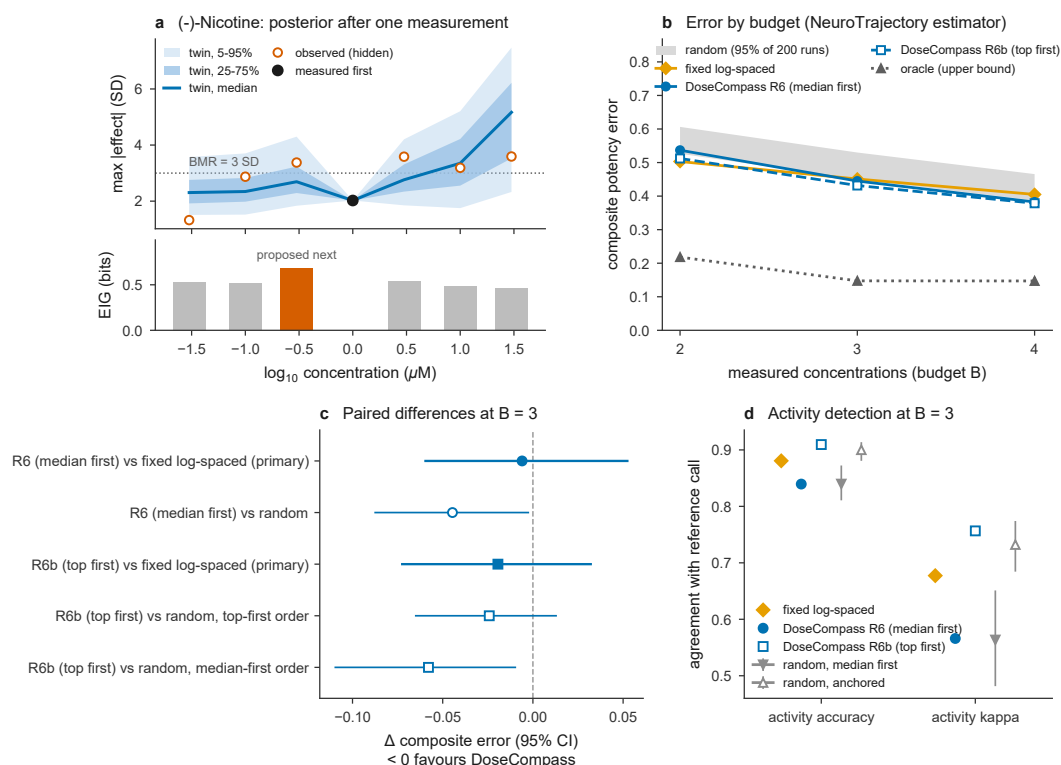


Figure 9. R6/R6b DoseCompass. (a) Twin posterior of the maximal developmental effect after one measurement of a reference-positive chemical, and the expected information gain of each candidate level. (b) Composite potency error against the budget B for each strategy, with the 2.5–97.5% band of 200 random orders. (c) Paired differences at $B = 3$ in the two registered protocols. (d) Activity detection at $B = 3$.

Table 10. R11 early exit from DIV 5/7 (nested 95%-sensitivity thresholds, $n = 194$). Sensitivity = active chemicals that continue to DIV 12; exit = inactive chemicals stopped at DIV 7; Clopper–Pearson 95% CIs; AUROC with chemical bootstrap CI.

Score (DIV 5/7 only)	Sensitivity [95% CI]	Inactive exit [95% CI]	AUROC [95% CI]
NeuroTrajectory (DIV 9/12 masked)	0.945 [0.89, 0.98]	0.458 [0.31, 0.61]	0.897 [0.85, 0.94]
NeuroTrajectory, upper 90% band	0.938 [0.89, 0.97]	0.417 [0.28, 0.57]	0.833 [0.76, 0.89]
Rule on DIV 5/7 data	0.945 [0.89, 0.98]	0.500 [0.35, 0.65]	0.920 [0.88, 0.95]
Persistence (DIV 7 carried forward)	0.952 [0.90, 0.98]	0.917 [0.80, 0.98]	0.987 [0.97, 1.00]

band 3 (11 against 1 discordant chemicals, exact McNemar $p = 0.006$; a secondary, post-hoc test), while both stop a similar number of the 15 negatives (8 and 7). The system therefore offers two gates: persistence for throughput, and the twin’s upper band when a missed neurotoxicant is the costlier error (15% of chemicals stop).

Reading. Most of the signal the complete assay will show is already present at DIV 7, which makes an early-exit step practical for a neural-OoC screening lab: with persistence, 26% of all chemicals stop after two of the four recordings. The circularity control and the external labels below qualify the choice of gate. The primary twin was never trained to extrapolate in time, which explains its lower efficiency.

R11b – retraining the twin to extrapolate in time. A second protocol, registered after R11 and before any training, changed one thing: during training, the context wells lose DIV 9/12 with probability one half, while the targets stay complete. Retraining raised the twin’s inactive exit rate to 69% (AUROC 0.955; paired gain over v1 0.23 [0.07, 0.39]), still below persistence (−0.23 [−0.36, −0.10]). Against the circularity control, a reference built from DIV 9/12 only (40 inactive chemicals), every early score loses ground and persistence still leads: it stops 48% of the inactive chemicals against 25% for the retrained twin, whose gain over v1 disappears there (−0.10 [−0.26, 0.05]). Retraining did not harm dose interpolation ($k = 3$ curve MAE −0.4%, [−1.3, 0.5]%), so the primary model remains the delivered forecaster. For a screening lab the recommendation is an auditable gate at DIV 7: persistence (the DIV 7 network carried forward stays below the activity threshold) when throughput

Table 11. R8: leave-one-recording-out errors (macro mean over 9 arrays) and the directed pair parameters fitted on all unstimulated arrays. Permuting directions leaves conduction times unchanged, so that column equals the model by construction. $d = (n^{\text{ff}} - n^{\text{fb}})/(n^{\text{ff}} + n^{\text{fb}})$.

Metric	Shrinkage model	Tunnel mean	Direction-permuted	Δ model – tunnel mean [95% CI]
Feed-forward fraction MAE	0.363	0.399	0.403	−0.036 [−0.056, −0.018]
Conduction-time MAE (ms)	0.154	0.161	0.154	−0.007 [−0.015, 0.001]
Conduction-time NLL (nats)	2.369	2.199	—	0.170 [−0.392, 0.992]
Directed pair	$n^{\text{ff}} / n^{\text{fb}}$	d	Mean conduction (ms)	
EC→DG	21 / 10	0.35	0.44	
DG→CA3	15 / 36	−0.41	0.39	
CA3→CA1	23 / 31	−0.15	0.49	
CA1→EC	30 / 30	0.00	0.43	

Table 12. R9: change in curve MAE when a component is removed (ablated minus full; positive = the component helps), paired over the 243 chemicals.

k	Full (seed 0)	No interpolation-informed decoder	No dose-local attention	Single seed instead of ensemble
1	1.432	0.032 [0.014, 0.051]	0.007 [−0.007, 0.020]	0.018 [0.011, 0.026]
2	1.294	0.057 [0.038, 0.076]	0.006 [−0.007, 0.020]	0.014 [0.006, 0.022]
3	1.187	0.082 [0.062, 0.104]	0.021 [0.008, 0.035]	0.006 [−0.002, 0.013]

matters most, or the twin’s upper band when a missed neurotoxicant is the costlier error; retraining the twin for the time dimension did not change that choice.

5.7 R8 — The chip layer on a real four-compartment MEA

Protocol. Leave-one-recording-out over the 9 unstimulated arrays of the Brewer data (Equation (7)); the stimulated recordings are described but not pooled, because their file identifiers do not map to physical arrays. The model predicts each held-out array’s per-tunnel feed-forward fraction and its axons’ conduction times, against per-tunnel means of the training arrays and a direction-permuted null. Integrity of the asset was checked first: the sign of the upstream–downstream spike lag agrees with the labelled direction for 437/437 measurable axons, and the absolute lag correlates with the conduction time at 0.984.

Result. Shrinkage by anatomical pair and tunnel predicts a new array’s directed connectivity better than per-tunnel means (Table 11 and Figure 10); against the direction-permuted null the gain (−0.040 [−0.096, 0.014]) is not significant with 9 arrays; every leave-one-recording-out comparison in Table 11 rests on 9 arrays and is underpowered. Conduction times are not predicted better than tunnel means, and the likelihood is not improved. Population bursts in the four compartments are associated overwhelmingly at zero lag: this is synchrony, not evidence of directed propagation, and we do not interpret it as conduction.

Reading. The chip layer is a small, honest component: calibrated descriptors of directed connectivity (feed-forward fractions with uncertainty, conduction-time distributions) that a compartmentalised-chip laboratory can compute on its own recordings. It is not calibrated across platforms and there is no drug in these data, so compartment dosing scenarios remain labelled simulations.

5.8 X2 and X3 — Cross-assay corroboration

Two unpaired comparisons with public assays, fitted on nothing external (Appendix C). Against neurite morphology imaging (X2, 55 chemicals), network function is altered at lower concentrations than morphology (median 0.56 $\log_{10}$ units) and the twin from three concentrations preserves the potency ranking. Against acute MEA exposure (X3), developmental activity predicts acute neuroactivity above chance, but potency ranks do not transfer.

5.9 R9 — Ablations

Each ablation removes one component of the v2 configuration chosen in each outer fold and retrains with the same seed and data; the reference is the full seed-0 model (Table 12, Figure 11a).

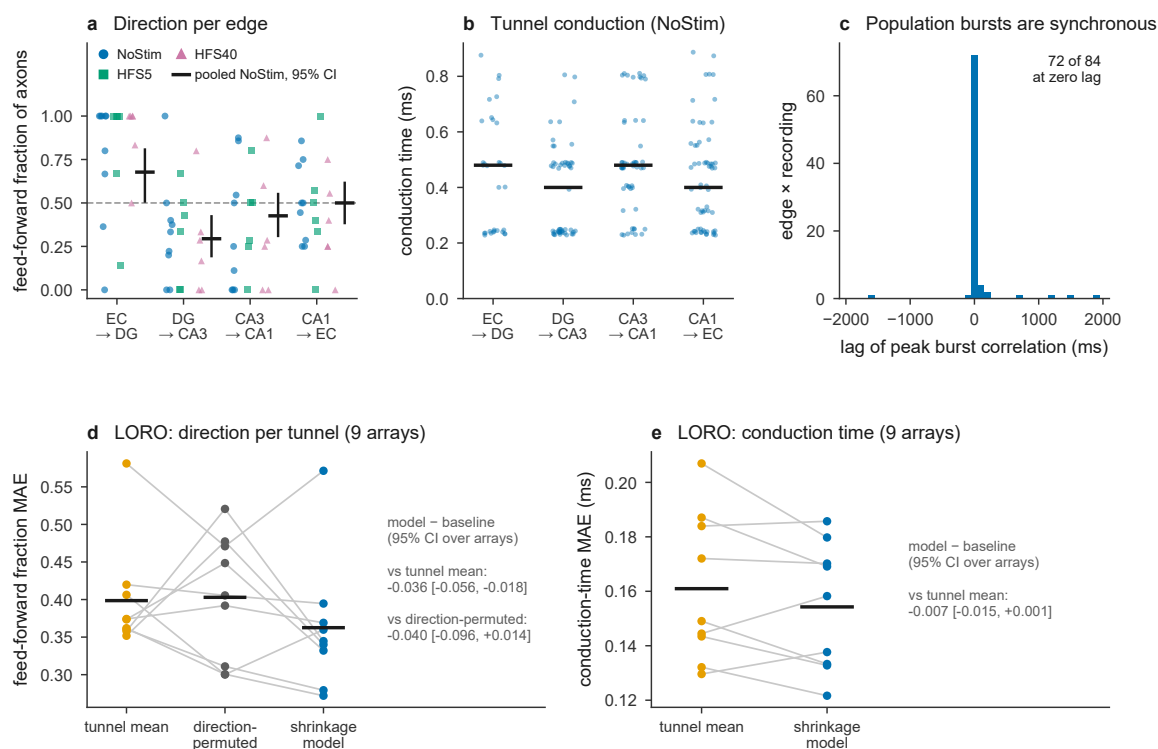


Figure 10. R8 chip layer (Brewer four-compartment MEA, CC0). (a) Feed-forward fraction of sorted axons per directed pair and recording. (b) Conduction times per pair. (c) Lag of the peak population-burst correlation between compartments: associations are synchronous. (d, e) Leave-one-recording-out errors of the shrinkage model and the baselines for each held-out array.

The interpolation-informed decoder matters at every k , and more as doses are added (+6.9% error without it at $k = 3$). Dose-local attention helps only once there are several doses to attend to (significant at $k = 3$ only); the three-seed ensemble helps most when the context is smallest. Each component earns its place where the design says it should.

5.10 R10 — Where the model fails

NeuroTrajectory improves on the best baseline for 77.3% of the never-inspected chemicals at $k = 3$; Table 13 and Figure 11(b, c) show where it does not. Error is highest for metals, whose effects are abrupt and cytotoxic, and the model loses to interpolation on pharmaceuticals and on the few GABAergic chemicals, whose responses are idiosyncratic in dose, and on food additives. It gains most on the large, mechanistically coherent pesticide classes (organochlorines, pyrethroids, acetylcholinesterase inhibitors), which are well represented in training. No network potency can be estimated for 60 chemicals, which never reach the benchmark response in the measured data; they are counted per class, not hidden. Class labels follow the EPA annotation file, which mixes two vocabularies; classes with fewer than five chemicals are shown only where they are informative, and no interval is claimed per class.

5.11 R5 — Specific versus cytotoxic disruption: a negative result

We asked whether a selectivity index—network potency relative to chemical-level cytotoxicity potency—separates DNT-positive from DNT-negative reference chemicals. It does not: AUROC 0.515 [0.349, 0.677], against 0.658 for network potency alone and 0.718 [0.613, 0.816] for cytotoxicity alone (Figure 11d). In this reference set cytotoxicity itself carries the label: 5% of the 19 negatives are cytotoxic, against 56% of the 79 positives. Separating specific from cytotoxic network effects needs same-well viability readouts, which the public NFA record does not provide. We therefore use cytotoxicity only to stratify calibration (R7), where it does help.

5.12 Is anything learnable left? A Darwin-Cage honest-ceiling audit

Question and protocol. After NeuroTrajectory, is there structure left in its out-of-fold errors that a broad automated search could exploit? We ported the team lead’s Darwin-Cage search (attribution in NOTICE). It

Table 13. R10: curve MAE at $k = 3$ by annotated chemical class (all folds; best baseline = log-linear interpolation). “Better” is the share of chemicals for which NeuroTrajectory beats the baseline; “n.e.” counts chemicals without an estimable network potency.

Class	n	NeuroTrajectory	Interpolation	Better (%)	n.e.
Organochlorine	15	1.340	1.779	93	0
Pyrethroid	16	1.113	1.360	100	0
Acetylcholinesterase inhibitor	27	1.137	1.374	74	8
PAH	16	0.827	0.909	94	7
Industrial	13	1.063	1.135	69	5
Pesticide (other)	11	1.115	1.191	64	5
Metal	9	1.819	1.934	56	2
Pharmaceutical	38	1.263	1.252	63	14
GABAergic	4	1.290	1.222	50	0
Unannotated	55	1.161	1.349	85	11

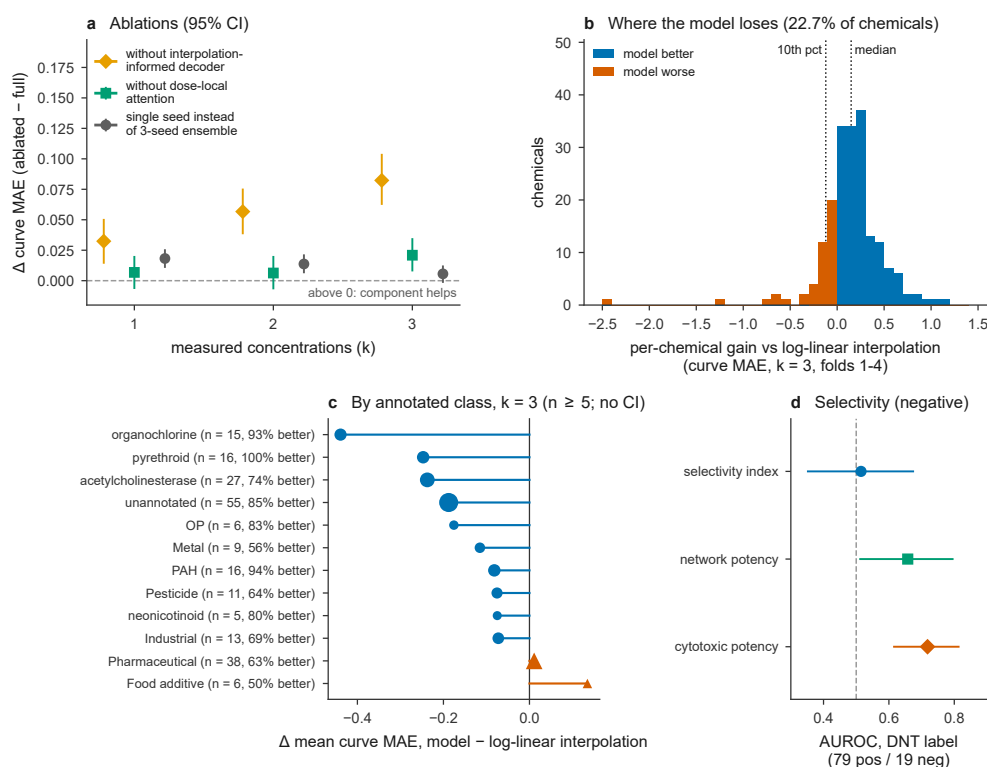


Figure 11. Diagnostics. (a) R9 ablations: change in curve MAE when a component is removed (95% intervals). (b) R10: distribution of the per-chemical gain over log-linear interpolation at $k = 3$, folds 1–4. (c) R10: mean gain by annotated class with at least five chemicals (no interval claimed). (d) R5: AUROC of the selectivity index and of its two components for the DNT reference label (negative result).

evolves small “programs” that group residuals by at most four atoms built only from information available at prediction time (feature, DIV, k , side of extrapolation, plate, class, and binned dose, distance to the nearest measured level, predicted value, predictive scale, vehicle level and replicate count). A program counts as confirmed only on chemicals the search never saw: it must pass a correlation threshold and a chemical-level sign-flip test (details in Appendix B). Planting structures of known size turns the search into a power analysis.

Result. The search finds real but unhelpful structure. On the primary confirmation fold it confirms 926 of 2007 evaluated programs (about 2 expected by chance), and almost all of them contain the model’s own predicted value: residuals shrink towards the mean, a regression-to-the-mean pattern. Most of them sit at the resolution floor of the sign-flip test, so the count measures breadth, not strength. Subtracting the best confirmed program, strictly nested, makes the forecast *worse* at every k (curve MAE 1.7% at $k = 1$, [0.010, 0.037] in absolute units; 2.2% at $k = 3$), and a generic gradient-boosting corrector with nested selection changes nothing (−0.4% to 0.6%, every interval includes zero). The audit would have found a plate shift of 0.4 vehicle

SD and a feature $\times$ DIV $\times$ class interaction of 0.64 vehicle SD with power 0.8.

The value of viability. One variable does carry information the model lacks: cytotoxicity. Adding the chemical-level AB/LDH viability calls cuts the curve MAE by 8.3% at $k = 1$ $[-0.145, -0.093]$ and by 5.5% at $k = 2$. Those calls, however, were measured on the same plates and summarise every concentration, including the hidden ones, so this is a *scenario*: what a prior viability screen would be worth, not an improvement of the forecast. Viability from the measured concentrations only, the leakage-free version, adds nothing (0.01% at $k = 1$ on the 51 chemicals where it exists).

Reading. Within the information available at prediction time, NeuroTrajectory sits at an honest ceiling: what remains is replicate noise and chemical-specific behaviour. For a sponsor, the actionable finding is concrete: a cheap viability read-out across the full concentration range, run before or alongside the network assay, is the one additional measurement that the audit shows would carry learnable signal.

6 Interpretability and reliability

Outputs in biological units. Every forecast is expressed in the same units as the laboratory's own quality control: standard deviations of the vehicle wells of the same plate and day (Equation (1)). A value of -3 at DIV 12 means the treated network sits three vehicle standard deviations below its controls, the threshold used for activity. Potency is reported as a concentration in μM with the feature that drives it.

Mechanisms that can be inspected. The decoder gate $\sigma(\gamma)$ (Equation (3)) can be read out per feature and day: it states how much of a forecast is interpolation of the measured wells and how much is learned correction; the attention weights of $r_t(q)$ state which measured concentrations inform a query. The predictive scale separates replicate noise (aleatoric) from disagreement between ensemble members (epistemic), and the epistemic ratio drives abstention. Calibration is reported per Mondrian stratum, so a user sees that cytotoxic chemicals receive wider intervals and why.

What we do not claim

- **Not human, not clinical.** All training data are rat primary cortical cultures (and rat hippocampal cultures for the chip layer). Nothing here predicts human developmental outcomes.
- **Not a validated twin of any commercial chip.** The public chip data come from rat hippocampal sub-networks without drug exposure; compartment-specific dosing scenarios are labelled simulations.
- **Coverage inside the calibration domain only.** Conformal coverage is measured on held-out NFA chemicals and is not transferred to other species, cell sources, assays or chips.
- **No superiority over EPA's rules.** With 19 negatives the hazard comparison is underpowered; one of three paired tests is significant, and EPA's Top- k rules were ranked in sample.
- **No model-derived potency.** It lost to interpolation (R3/R3b); BMC is reported by interpolation of measured doses.
- **No causal propagation.** Zero-lag burst associations between compartments are synchrony; conduction times come from sorted axons, not from burst lags.
- **No validated feature engine.** R1 was not estimable; the twin uses EPA's published features.
- **Retrospective planning.** DoseCompass was validated by revealing wells that had already been recorded, on EPA's concentration grid.
- **Viability does not improve the forecast yet.** Its gain in Section 5.12 is a scenario that uses cytotoxicity measured at the hidden concentrations; viability at the measured concentrations adds nothing.
- **Early exit has two gates, not one winner.** Persistence maximises exits against the assay's own rule; the twin's upper band loses fewer known neurotoxicants (a secondary, post-hoc comparison).

7 Practical value and impact

Wells, recordings and days. Most chemicals (212 of 243) were tested at seven concentrations. Measuring three of them, a laboratory uses 57.1% fewer treated wells per chemical (median) and still obtains (i) the full dose $\times$ DIV trajectory of all 17 features with calibrated intervals, (ii) a hazard call that detects more known neurotoxicants than the published rules on the full series, and (iii) a potency by interpolation of the measured doses, with the uncertainty of the twin shown alongside. On a 48-well plate the saved wells become more compounds per plate, or more replicates where the twin abstains. The DIV 7 gate (R11) stops 26% of chemicals

Table 14. Roadmap for a neural organ-on-a-chip laboratory (proposed, not executed).

Step	What	Success criterion
1. Asset mapping	export chip MEA recordings to AssaySpec/ChipSpec; compute the 17 features per compartment	provenance and licence per row; features reproduce the laboratory's own summaries
2. Few-shot adaptation	meta-fine-tune NeuroTrajectory on the sponsor's compounds with chemical-level folds	held-out curve error below per-compound interpolation, as in R2
3. Chip-specific calibration	Mondrian conformal strata by chip design and compartment	observed coverage at nominal on held-out compounds
4. Directed connectivity	add compartment-resolved outputs (feed-forward fraction, conduction) to the forecast	leave-one-chip-out gain over per-tunnel means, as in R8
5. Prospective planning	randomise compounds to DoseCompass or fixed designs on a free concentration range	fewer wells to the same potency and hazard precision

(persistence) or 15% (the twin's safety gate) after two of four recordings, saving two recordings and five of twelve culture days for each; this applies directly to single-condition chips and to plates whose layout groups chemicals. Planning from DIV 5–9 loses nothing (R6), so the next concentration can be chosen before the last recording. The next assay worth adding is a full-range viability read-out (Section 5.12).

A workflow for a neural organ-on-a-chip laboratory.

1. *Describe* the experiment in the AssaySpec/ChipSpec schema (compound, concentrations, days, compartments, tunnel directions); recordings enter the provenance-tracked asset.
2. *Anchor*: measure the top concentration first (R6b), then the concentrations DoseCompass proposes.
3. *Forecast* every untested concentration and day with NeuroTrajectory, with conformal intervals; read the abstention flag.
4. *Stop early*: at DIV 7, apply the throughput or the safety gate (R11), chosen by the cost of a missed neurotoxicant.
5. *Decide*: hazard call; BMC by interpolation of measured doses; if the twin abstains, add a concentration.
6. *Describe the chip*: feed-forward fractions and conduction-time distributions per tunnel and pair, with uncertainty, from axon-sorted recordings (R8).
7. *Update*: each finished experiment joins the training asset and the twin is retrained in batch (no online assimilation yet); the next compound starts from a better prior.

Roadmap with sponsor data. The same code runs on a sponsor's recordings once they are mapped to the schema. Table 14 lists the steps we propose and how each would be judged; none of them is claimed as done.

Impact. Developmental neurotoxicity is among the most expensive endpoints to test in animals, and the in-vitro battery is entering regulatory use [1, 11]. A calibrated twin that states when it can be trusted, and that turns each new plate into an update of a shared prior, lowers the cost per compound, reduces animal testing in follow-up studies and gives chip developers a way to make their recordings a reusable, predictive asset.

8 Reproducibility

- **Frozen inputs.** Cohort, aliases, quality rules, normalisation and splits are frozen by SHA-256 (freeze G0, payload b77e34100f9bbe8e724551075c339dbc10d9a9e170233d210cf4d5b505d02ef7); a checker verifies them against their hashes, and the one amendment is documented (Appendix D).
- **Registered protocols.** R6, R6b, R11 and R11b were written to protocol files and hashed before any result was computed; R3b was registered in the team's decision log before running. Post-hoc secondary analyses requested by our internal reviews are in report_secondary.json and labelled as such. Development runs that were seen before a final run are declared inside each result file.
- **One source of numbers.** Every number in this report, the README and the Writeup is rendered from

results/*.json and listed in a ledger; rendering stops on an unresolved token, and CI rejects untraced numbers in every rendered document, this report included.

- **Runs anywhere.** `pip install -r requirements.txt` && `python demo.py` reproduces a held-out forecast on CPU without network access; the Kaggle notebook reproduces the headline tables; the cross-validated weights ship in `data_bundle/`. Full retraining uses the scripts of Table 3 with fixed seeds.
- **Tests.** A pytest suite covers ingestion, the cohort freeze, features, trajectories, the planner, the chip layer, the cross-assay joins and the app core.
- **Licences.** Code MIT; dependencies under permissive licences (PyTorch, NumPy, SciPy, pandas, scikit-learn, h5py: BSD; rdata, openpyxl: MIT; pyarrow, Gradio: Apache-2.0); the AGPL-licensed pyreadr was removed. Data licences are in Appendix E.

9 Limitations

- **Species and system.** Rat cortical cultures on a single assay platform; transfer to human iPSC-derived networks or to the sponsor's chips is untested.
- **Small negative class.** 19 DNT-negative reference chemicals limit every specificity estimate.
- **Chip data.** 9 unstimulated arrays, no pharmacology and no mapping between stimulated recordings and physical arrays; conduction times are quantised by the sampling rate.
- **Secondary splits.** Plate, dose-level and chemical-family splits are frozen but not yet evaluated.
- **Baselines.** Comparators are statistical and nearest-neighbour models, EPA's own rules and the team lead's earlier architectures; no descriptor-based QSAR model or Gaussian-process dose-response model was run.
- **Multiplicity.** Dozens of intervals and tests are reported without correction; primary endpoints are marked as such, and post-hoc secondary analyses are labelled.
- **Registration anchor.** Protocols were hashed before their results, sometimes only minutes before, on the same machine; the freeze predates the first git commit. The public release anchors both to a tagged commit.

10 Prior work, AI tools, team composition and licences

Prior work. A first version, NeuroChip Twin v1 (github.com/Agnuxo1/neurochip-twin, tag v1-synthetic, MIT licence, 2026-09-21), was a synthetic time-lapse phenotyping prototype. Version 2 replaces its synthetic benchmark entirely with real neural electrophysiology; no model, data or result of that prototype is reused. (Elsewhere, “v1” denotes the primary NeuroTrajectory configuration of this submission.)

AI tools. The system was designed and built by the team lead with two AI coding assistants working as a pair through an auditable gateway that records every exchange, decision and file lock: Claude Code (Anthropic; model Claude Opus 5.5) and OpenAI Codex (model GPT-6-Sol). Each proposed an independent strategy; the plans were cross-reviewed and merged; every module had one owner and a designated reviewer from the other assistant. Where the second assistant's review is still pending, analyses were re-run by an independent verification agent that recomputed every reported number from the result files; pending cross-reviews are listed in the decision log. The team lead verified the results and takes responsibility for every claim. No generative model is part of the delivered scientific pipeline.

Team composition (AI + biology). Francisco Angulo de Lafuente (ORCID 0009-0001-1634-7063), the sole human team member, combines applied microbial biotechnology and AI research. *Biology:* sole inventor of two Spanish patent families on the bacterial and fungal fermentation of organic waste into fuels (CPC C12P; ES2273594B1, ES2341194B1, with WO2009034217A1), co-inventor of two process-plant patents (ES2402644B1, ES2438092B1) and co-author of the book *ECOFA* (2008, ISBN 978-84-936176-4-6). *AI:* four arXiv preprints (arXiv:2601.01916, 2601.09557, 2601.12032, 2604.19792), a journal article on physical reservoir computing (doi:10.5958/0975-8089.2025.00003.0) and open-source software including NeuroChip Twin v1. His biology is microbial bioprocessing rather than neuroscience; the neuro-specific design follows the cited EPA and Brewer protocols. Claude Code and Codex are declared as tools, not team members.

Licences. Code MIT; data as in Appendix E; no non-commercial dataset and no code without an OSI licence.

Table 15. Where each evaluation criterion is addressed. Weights are those announced for the Kaggle and Pazhou evaluations.

Sections	Kaggle criterion (weight)	Pazhou criterion (weight)
§1 Problem; §7 Practical value	Impact and importance (30)	Practical value (实际价值) (20)
§3 Method; §4 System	Technical approach, innovation (30)	Technical innovation (技术创新性) (30)
§5 Results, protocols and negatives	Results and validation (20)	Completion and effect (完成度与效果) (25)
§6 Interpretability; §9 Limitations	Results and validation (20)	Interpretability, credibility (可解释性与可信度) (10)
§2 Data; §4 System; §8; App. D–E	Reproducibility (10)	Completeness (方案完整性) (15)
Figures, demo, video; §10 Disclosure	Presentation (10)	AI + biology team bonus (+0.5)

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A Own prior architectures evaluated on real data

Why. Before choosing NeuroTrajectory (a recorded design decision) we tested three of the team lead’s earlier, published architectures on the same real data. We applied the rule used everywhere else in this report: an own architecture enters the system only if it beats the baseline under the same split and metric. None did, and no component of this appendix is used by the headline model. We report them because the same rigour applies to our own ideas as to anyone else’s.

Candidates (described by what the code computes). (A) An *eikonal propagation layer* on the chip geometry: constant-speed arrival times on the device mask (walls and the 20 tunnels as on the chip), fitted to the median network-burst latency of held-out electrodes of the Brewer MEA; from Agnuxo1/Optical-Neuromorphic-Computing-for-Real-Time-Pathfinding (MIT, Zenodo 17583729). Burst latency correlates with firing rate, so it is not called a propagation velocity. (B) A classical, Schrödinger-inspired *lattice reservoir* on the chip geometry with a linear readout, corrected to an exactly unitary Crank–Nicolson step; nothing quantum is computed. It forecasts square-root spike counts of every well electrode one and four time bins ahead ($h = 1$, $h = 4$); from Agnuxo1/QESN_MABe_V2_REPO (Apache-2.0, Zenodo 17266084). (C) *DoseLattice*: the same repository’s mass-conserving diffusion stencil in one dimension over $\log_{10}$ concentration, used as a fixed encoder of the context wells inside NeuroTrajectory. It was evaluated on the development fold 0 only (49 chemicals), never on the reported folds 1–4.

Table 16. Own prior architectures on real data: paired differences, candidate minus comparator, with 95% bootstrap CIs (A, B: over the 9 Brewer recordings without stimulation; C: over chemicals). Negative is better for the candidate.

Candidate	Comparator	Metric	Δ [95% CI]	Verdict
A eikonal	IDW within compartment	latency MAE (ms)	0.21 [−0.21, 0.65]	no gain over interpolation
	same, no walls (ablation)	latency MAE (ms)	−0.21 [−1.57, 1.01]	geometry adds nothing
B lattice, $h = 1$	ridge VAR	test MSE (%)	2.1 [−2.4, 6.3]	matches, does not beat
	echo-state network	test MSE (%)	1.3 [−2.9, 5.6]	matches, does not beat
B lattice, $h = 4$	ridge VAR	test MSE (%)	−0.8 [−6.7, 3.1]	matches, does not beat
	echo-state network	test MSE (%)	−1.8 [−7.5, 3.0]	matches, does not beat
	electrodes shuffled (ablation)	test MSE (%)	1.9 [0.4, 3.6]	real geometry slightly worse
C DoseLattice, $k = 1$	NeuroTrajectory v1	curve MAE (%)	0.6 [−0.9, 2.0]	no gain
C DoseLattice, $k = 4$	NeuroTrajectory v1	curve MAE (%)	3.7 [0.9, 7.5]	worse

Reading. (A) and (B) match simple baselines but do not beat them, and their chip-geometry ablations show that the geometry itself carries no measurable signal at this recording scale. (C) adds redundancy to a model that already interpolates in log dose and attends locally in dose, and it gets worse as more concentrations are measured. The ideas that did transfer are conceptual: the calibrated early-rejection principle behind R11 (Section 5.6) and the honest-ceiling residual audit of Section 5.12. A larger chip with stimulation protocols, where propagation is measured rather than inferred from bursts, is where (A) and (B) could be tested again.

B Darwin-Cage honest-ceiling audit: details

Search space and budget. Residuals are ensemble-mean predictions minus observed level means for the 194 chemicals of folds 1–4 and $k = 1, \dots, 4$ (1,124,784 rows; they reproduce the curve MAE of trajectory_cv.json to within 5e-05). A program is a composition of up to four atoms: a categorical column, or a numeric column that is scaled, divided or quantile-binned. Its fitness is the out-of-fold correlation between the shrunken group mean and the residual, rotating over the three search folds. The search evaluates about 2007 programs per rotation. A program is confirmed on the held-out fold if its correlation exceeds $z/\sqrt{n}$ with $z = 3$ and a chemical-level sign-flip test gives $p < P(Z > z)$; each fold is used once as the confirmation fold (sensitivity analysis).

Power. Structures of known size (a plate | date shift, and a feature $\times$ DIV $\times$ class interaction) were planted in the residuals at 0.05–0.8 vehicle SD, 5 replicates each. *Exact* recovery means that the search confirms a program containing every planted atom. The minimum detectable effects at power 0.8 are 0.4 and 0.64 vehicle SD; handing the planted program directly to the confirmation test lowers the second to 0.35. Small

plants are always “found” in the loose sense, because the real residuals already contain confirmable structure; this is why exact recovery is the primary criterion.

Viability arms. Four arms are compared with the same nested, chemical-grouped corrector (identity or gradient boosting, chosen on the training folds): prediction-time covariates only (V0); plus PubChem AB/LDH calls (V1, scenario); plus NTP DNT-DIVER per-well viability summarised over all concentrations (V2, scenario); plus NTP viability at the measured concentrations only (V3, no hidden-concentration information); and a control arm with only an indicator of NTP availability. The NTP file is not an independent experiment: for 60 of its 64 chemicals the concentration sets are identical to the NFA wells, so it is treated as a same-plate replication.

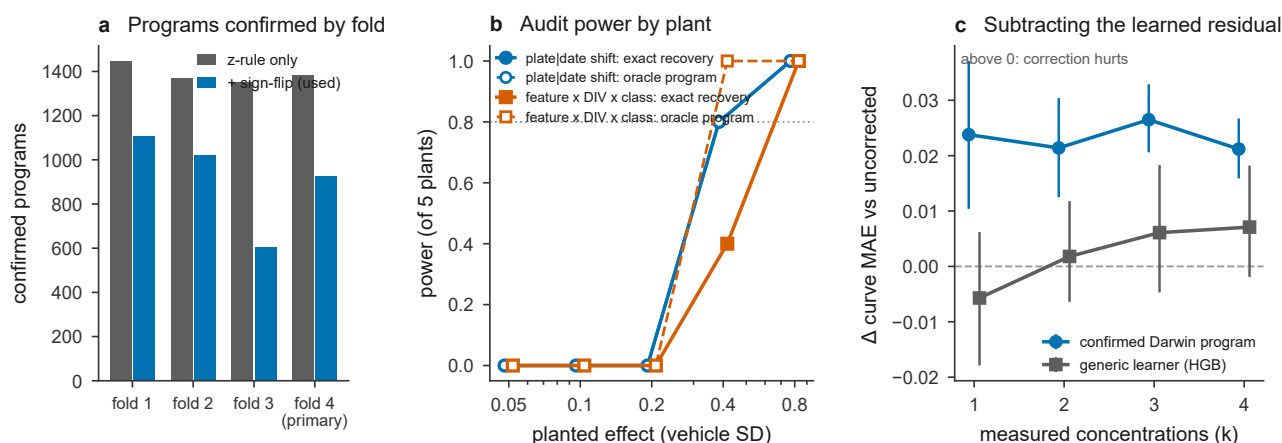


Figure 12. Darwin-Cage audit. (a) Programs confirmed on unseen chemicals for each confirmation fold. (b) Power: fraction of five planted structures recovered, by the search and by the planted program itself. (c) Curve-MAE change when the learned residual structure is subtracted (nested), for the best confirmed program and for a generic learner, with 95% bootstrap CIs.

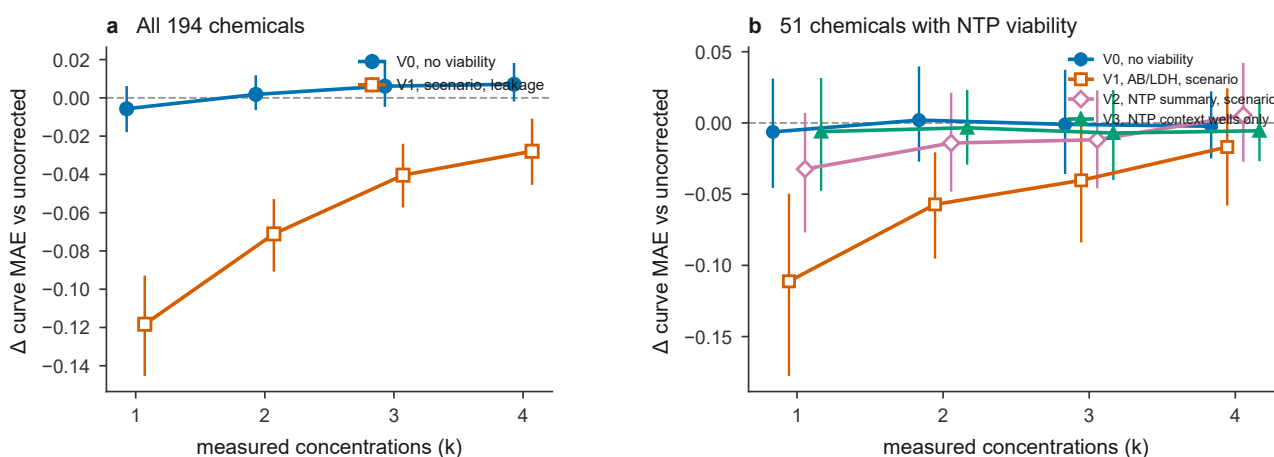


Figure 13. Value of information of viability (nested corrector, curve-MAE change with 95% bootstrap CIs). (a) All chemicals: prediction-time covariates only (V0) and plus AB/LDH calls (V1, a scenario with leakage). (b) Chemicals with NTP per-well viability: V2 (all concentrations, scenario) and V3 (measured concentrations only).

C Cross-assay corroboration (X2, X3): details

Both analyses are *unpaired*: different plates, cultures, exposure windows and readouts. No parameter is fitted on the external data.

X2: network formation versus neurite morphology (Harrill imaging). On the 55 chemicals matched by exact name or synonym, a chemical that alters neurite or synapse morphology is almost always active in the NFA ($P = 0.958$), whereas many NFA-active chemicals leave morphology unchanged; activity agreement is 0.62 [0.49, 0.75] ($\kappa = 0.29$). Among the 23 chemicals active in both, potencies correlate (Spearman 0.50

[0.006, 0.854]) and the network BMC is lower by a median of 0.56 $\log_{10}$ units [−0.98, −0.37]: network function is altered at lower concentrations than morphology (Figure 14a). The NeuroTrajectory forecast from three concentrations preserves this ranking (Spearman 0.72 [0.35, 0.91]), although it calls more chemicals active than the morphology assay. The direction of change agrees no more often than expected by chance (0.59 vs 0.58).

X3: developmental versus acute exposure (Kosnik). On 99 overlapping chemicals, NFA developmental activity predicts the published acute-MEA call above chance (balanced accuracy 0.654 [0.561, 0.750]; the twin from three concentrations: 0.630), with high sensitivity but low specificity (0.38). Potency ranks do not transfer (Spearman 0.17 [−0.081, 0.415], one-sided $p = 0.084$). The literature truth set overlaps the cohort in 30 neuroactive but only 3 negative chemicals and is not interpreted. Acute neuroactivity and developmental disruption are different questions; the agreement corroborates shared neuroactive chemistry, not a DNT prediction.

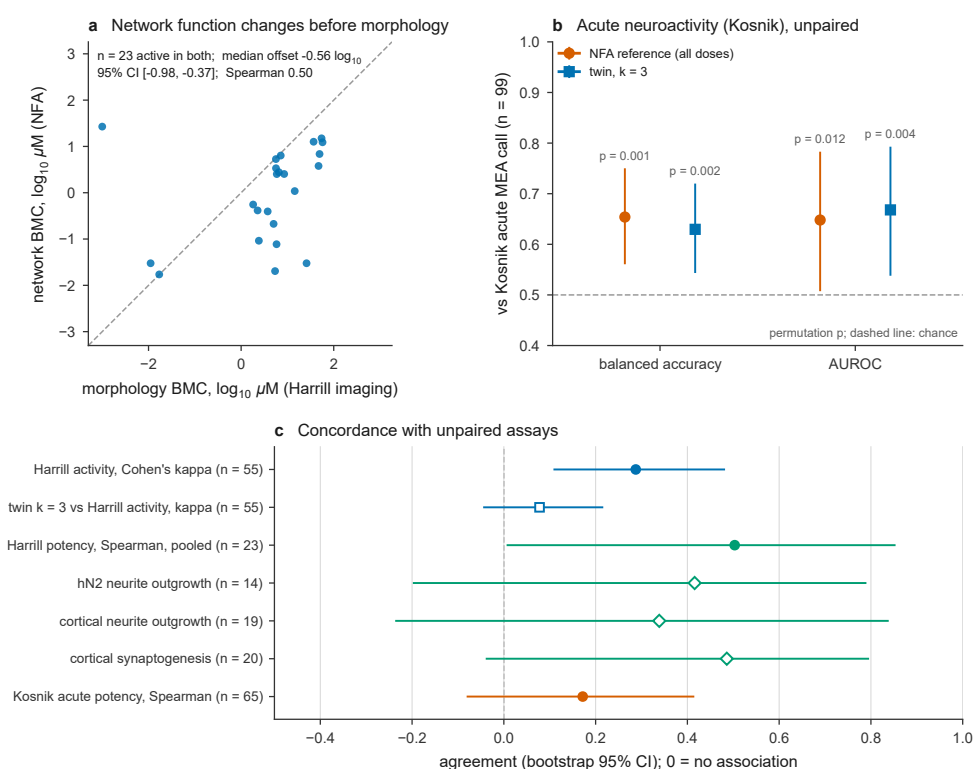


Figure 14. X2/X3 cross-assay corroboration (unpaired). (a) Network (NFA) versus morphology (Harrill imaging) benchmark concentrations for chemicals active in both; points below the diagonal change network function at a lower dose. (b) Balanced accuracy and AUROC of the NFA reference call and of the twin for the published acute-MEA activity call. (c) Agreement statistics with 95% intervals.

D Frozen inputs and protocol hashes

All registrations and runs took place on the same day and machine; some runs started minutes after their protocol was hashed (inference-only analyses are fast). The hashes prove that a protocol file did not change after its results were computed, not when it was written; the tagged public release anchors every file to a commit.

Table 17. SHA-256 fingerprints of the frozen inputs and registered protocols. Registration and run times are UTC.

Item	SHA-256 / identifier
Freeze G0 (freeze_g0.json), payload	b77e34100f9bbe8e724551075c339dbc10d9a9e170233d210cf4d5b505d02ef7
freeze identifier	G0-2026-09-26
primary chemical split (splits_epa_nfa.json)	3c0108201020ce055e3c92f4c574c601d96048b34e6d3880d74cd61f51bf2c45
cohort component	97110e816c39136c3a3a181baea94018d9467c2e853dfbc467fff8d14128d63
alias map	51692bc6b4165a2614d7d032cd64b8dfd57dd75fb8f188108d9bae058af27b95
quality rules	46cf3a2b6a01a7ad1cff8ecd78a646e7457897b41665d97ebbcf8b176194ee61
normalisation	756ce126b5520e36cb87c09a3050e770ca094efddf0e73b69cb05cda101d5782
EPA NFA source file (All_DIV_Data.Rdata)	fe8015c789770092b485fb63dd8aa0f643120e366df73c5624665454ee982af6
Amendment G0-A001 (pyreadr → rdata; no data or split change)	2026-09-26
R6 DoseCompass protocol	95c699a3ba9ead781c944de269db362b5a56ad301e4014662f5483cadeaa8495
registered / run started	2026-09-26T07:41:11+00:00 / 2026-09-26T08:57:18+00:00
R6b DoseCompass-anchored protocol	acf47e566f137680485258259b3d8691a8c78d60a8fb5eb7f01340d354d7817
implementation details	af4e5993cacea9fcd9f87185327e60339de2883575bfad62c33f5d2e71b6cd2
registered / run started	2026-09-26T09:43:10+00:00 / 2026-09-26T09:45:37+00:00
R6 result file, unchanged during the R6b run	4ef5a1fb1670310dc9fa9c444872568b3542c8ab7948b6bfa0b3f284c2a49967
R11 early-exit protocol	c92b9476be82b0a8b2d29556e096047edc8ff314505b4658b7f02c5ecabbf159
registered / computed	2026-09-26T13:03:27Z / 2026-09-26T13:05:00+00:00
R11b retrained-twin protocol	3ed9b54de673ee23f0a7f5afdc1d88ed87069e76ca052875a634eb48d6eb1a51
registered / computed	2026-09-26T13:15:00Z / 2026-09-26T14:29:41+00:00
R3b completed-series estimator	registered in the team decision log (2026-09-26, before execution)
EPA NFA analysis code (cited, not copied)	commit 01adf3e1a0068c87fe221d60df36b9f96c4b4b1d

E Data licence inventory

Table 18. Data used, where it comes from, its terms, and what the public repository redistributes.

Dataset	Source	Terms	In the repository
EPA NFA (All_DIV_Data)	US EPA, CompTox-DNT-NFA-Refinement (pinned commit, Table 17)	US Government work	derived bundle (normalised tasks, weights); raw file by download script
EPA DNT reference lists	same repository; [12, 13]	US Government work	labels used in evaluation
EPA tcplfit2 NFA fits	same repository	US Government work	not redistributed
PubChem AIDs 2284083, 2284068	PubChem [14] (US EPA deposit)	US Government work	chemical-level calls
NTP DNT-DIVER	CEBS, doi:10.22427/NTP-DATA-002-00062-0001-0000-1	US Government work	not redistributed
Brewer 4-compartment MEA	Dryad/Zenodo, doi:10.5061/dryad.7h44j1013	CC0 1.0	derived tables; raw files by checksum-verified download
Harrill DNT imaging	EPA ScienceHub, doi:10.23719/1407642	US Government work	not redistributed
Kosnik acute MEA	EPA ScienceHub, doi:10.23719/1504294	US Government work	not redistributed

Evaluated and not used. Datasets with non-commercial terms (for example LIVECell and the Cellpose training set); a public human iPSC network-formation set with too few compounds in common with the NFA cohort for a quantitative probe; a published neurotoxicity table whose host blocked automated download (not circumvented); multi-terabyte imaging collections outside the scope of a network-formation twin; and the public NFA spike archive, which shares no plate with the NFA record (R1, Section 2). The EPA analysis code has no licence file and is cited, not copied.